

TECHNICAL ASSISTANCE GUIDELINES



Course 291 ***AVIATION ENGLISH FOR*** ***AIR TRAFFIC*** ***CONTROLLERS***

For application by Instructors at ICAO-assisted Training Centres and for the unrestricted use of other interested Organizations

ENGLISH FOR AIR TRAFFIC CONTROLLERS

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INTRODUCTION

This Guideline has been prepared by the Technical Assistance Bureau as a language course primarily for would-be air traffic controllers for whom English is a foreign language. However, all personnel in contact with air traffic control, such as pilots, aerodrome firefighters, aerodrome maintenance personnel, etc. would also benefit from taking this course if English is not their mother tongue.

This course is confined to teaching the technical terms used in air traffic control. Before taking this course, therefore, students must have studied English to an advanced level. For a broader and more basic aviation vocabulary, they should have studied Supplements 1 and 2 to Technical Assistance Guideline No. 2 - Basic Aviation English and Technical Aviation English.

The direct method of teaching, or the aural-oral method, as it is also called, has been recognized all over the world for its effectiveness in making the language alive. The basic principles of the direct method of teaching are:

1. Direct association of perception and thought with the foreign speech and sound.
2. Constant and exclusive use of the foreign language.

Technical vocabulary is frequently explained with simple definitions. However, an explanation alone is not sufficient; practice in the use of the new vocabulary must be given in various ways so that the student can use it with ease in its affirmative, negative and interrogative forms.

It is intended that the lessons in English for ATC take about 30 hours to cover. Lessons are divided according to the subject matter and not according to the time required for their presentation. The instructor may give more than one lesson in an hour if the lessons are short and easy, or he may devote several hours to one lesson if necessary.

HINTS TO INSTRUCTORS

Each lesson should start with a revision of the previous lesson.

Before starting any new lesson, the instructor should be perfectly familiar with the contents of that lesson. Student participation can be encouraged by the use of audio visual aids, word games, and additional conversation. This guideline gives some ideas for conversation but they are by no means exhaustive.

To promote conversation, the instructor introduces the new vocabulary into the lesson by asking questions, which require affirmative answers. Then he asks questions requiring negative replies. After that, questions are asked in many different ways, using why, how, what, who, etc., until the students have sufficient practice. Ridiculous or improbable type of questions requiring negative replies is used not just for a laugh, but to assure the instructor that the student has understood the meaning. Without negative replies it is very likely that the meaning of the word has not been understood.

It is not always easy to understand the meaning with only one or two associations, particularly when it is introduced in relation to such technical terms that the student is not too familiar with, even in his own language. Always try to introduce a new word in a sentence that is personal if at all possible. An example can be the verb, "to move" (movable parts). If the student, because he has not understood, is feeling uncomfortable when you ask him such a question, you can change from the technical context. "Do you move in your chair when you are nervous?" Such familiar and directly personal associations leave no doubt as to what the meaning of the word is and, further, imprints the new word into his memory.

Remember, even if the meaning is quickly understood because of the similarity sometimes to the student's language, practice in pronunciation and its application in the English language is always required.

There is a tendency on the part of instructors to ask all the questions but since the sentence construction in the interrogative is different and sometimes difficult for students, there should always be some time set aside for students to ask one another questions.

When correcting a mistake, the instructor should merely pronounce the correct expression without quoting the mistake. Repeating the mistake may accustom the student's ear to the wrong expression.

The part of each lesson listing the vocabulary is included for easy and quick reference. No attempt has been made to give the various different meanings to each new word. It is not meant to be a dictionary; the explanations are not taken from dictionaries or lexicons, and other meanings applicable to the word have been ignored. It is merely an explanation of the meaning of the word or phrase within the context of the lesson. The conversation section of the lesson eliminates ambiguity.

The instructor might feel that more reading and writing should be covered in the course but remember, speaking is learned only by speaking, reading by reading, and writing by writing.

LESSON 1

FLIGHT BA 624 FROM MIRABEL INTERNATIONAL AIRPORT, MONTREAL TO HEATHROW INTERNATIONAL AIRPORT, LONDON

"Ladies and gentlemen, this is your flight attendant (1) speaking. I wish to welcome you aboard (2) British Airways flight 624 to London. Please fasten (3) your seat belts and observe (4) the "no smoking" signs. We wish you a comfortable and pleasant journey".

These, or similar words, along with emergency procedures, (5) are spoken countless times each day by flight attendants while the captain and his crew on the flight deck await taxi and take-off instructions from air traffic control.

While the passengers are getting ready for take-off and making themselves comfortable, quite unaware of what is going on (6) on the flight deck, the pilot has tuned in to (7) the ground control frequency and requested taxi instructions and route clearance (8).

Pilot: Mirabel ground control British Airways six two fower at ramp (9) two zero, request taxi.

Controller: British Airways six two fower Mirabel ground control. Taxi to runway tree zero via taxiway R two and W one.

Pilot: British Airways six two fower. Roger, to runway tree zero via taxiway R two and W one.

When the ground controller has BA 624's route clearance, he relays (12) it to the pilot.

Controller: British Airways six two fower. Route clearance.

Pilot: British Airways six two fower. Go ahead. (13)

Controller: British Airways six two fower. Cleared to Londor
Maintain flight level (14, 15) tree fife zero, squawk 4532.

Pilot: British Airways six two fower. Cleared to London.

Maintain flight level tree fife zero, squawk 4532.

When flight BA 624 reaches the holding point clear of runway (16, 17) 30 and is ready for take-off, the pilot contacts the local controller in the tower cab.

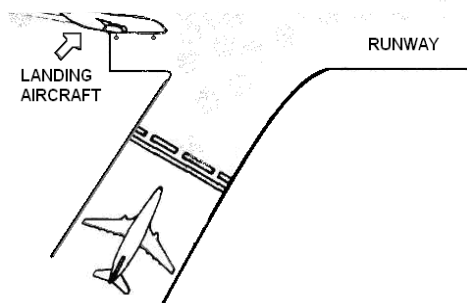
Controller: British Airways six two fower. After departure continue runway heading. (18) Cleared for take-off.

Pilot: British Airways six two fower. After departure continue runway heading. Cleared for take-off.

Controller: British Airways six two fower. Contact (19) approach control one two fower point six now.

Pilot: One two fower point six. British Airways six two fower.

BA 624 AT HOLDING POSITION



VOCABULARY

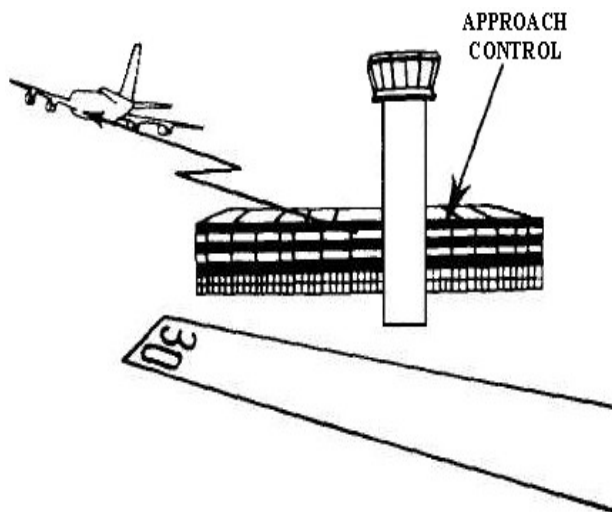
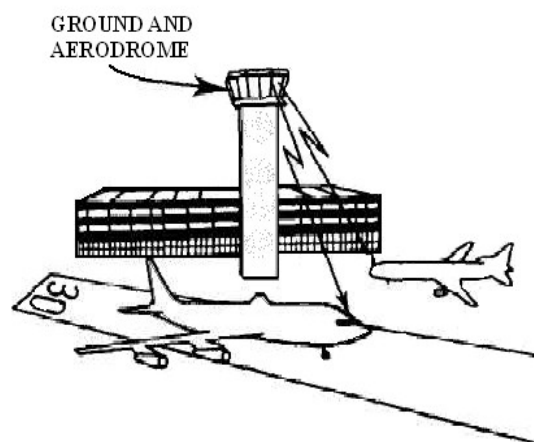
- | | |
|--------------------------------|---|
| 1. Flight attendant | - Stewardess or steward. |
| 2. Aboard | - On board or on an aeroplane, bus, train, ship, etc. |
| 3. To fasten | - To attach; to make secure. |
| 4. To observe | - To take notice of. The opposite is to ignore. |
| 5. Emergency procedures | - Safety procedures learned in case of an emergency. |
| 6. Unaware of what is going on | - Not knowing what is going on. |
| 7. To tune in to | - To choose a frequency (station) on the radio. |
| 8. Clearance | - Permission from ATC to taxi, take-off, land, etc. |

- | | |
|---------------------|---|
| 9. Ramp | - Another word for apron. An area on an aerodrome for loading and unloading passengers, cargo, refuelling, parking etc. |
| 10. Over | - Message is finished and I expect a response from you. |
| 11. Roger | - Message is understood. |
| 12. To relay | - To pass on (information in this case). |
| 13. Go ahead | - Pass your message, I am ready to receive (copy) your clearance. |
| 14. To maintain | - To fly (to proceed) or to keep (remain) at a flight level. |
| 15. Flight level | - Level (or altitude) which is maintained during a flight. |
| 16. Holding point | - Location where aircraft wait for line up and take off clearance. |
| 17. Clear of runway | - Not on the runway; a short distance from the runway. |
| 18. Heading | - Direction in which the aircraft is pointed; usually expressed in degrees from North. |
| 19. To contact | - To get in touch with; communicate with, to establish contact with, to change to another frequency. |

CONVERSATION

1. Does a flight attendant speak to the passengers? Does a flight attendant take care of the passengers? Does a flight attendant work in the galley? Is a flight attendant a mechanic? What does a flight attendant do? To whom does a flight attendant speak when the plane is taxiing? What is another name for a flight attendant?

2. Does a flight attendant welcome passengers aboard the aeroplane? Does he welcome the crew aboard? Does he welcome air traffic controllers aboard? Who does he welcome aboard? How does he address the passengers? What does he say when he welcomes passengers aboard? Who is aboard BA 624?



3. I close the door but I also fasten the door. I fasten this knot. Will you fasten your jacket please? Will you fasten your belt? What are you fastening? What are passengers asked to fasten? Why are they asked to fasten their seatbelts?

4. Are passengers asked to observe the "no smoking" signs? Who asks the passengers to observe the "no smoking" signs? When are they asked to observe the "no smoking" signs? Why are they asked to observe these signs? What is the opposite of observe? What do passengers observe when the plane is taxiing? What would happen if the passengers did not observe the "no smoking" signs?

5. While the plane is taxiing the flight attendant demonstrates to the passengers the emergency procedures. Are emergency procedures to be followed in case of engine trouble? Do we have to follow emergency procedures in case of fire?

When does a flight attendant demonstrate emergency procedures? Do we follow emergency procedures on every flight? When do we follow emergency procedures on a flight? At this school?

6. Are the passengers unaware of what is going on on the flight deck? Are flight attendants unaware of what is going on on the flight deck? Who is unaware of what is going on on the flight deck? Why are you unaware of what is going on in the next room?

7. I turn on the radio; I then tune in to such and such a station. Can a pilot tune in to ground control? On what frequency does he tune in to ground control? Can he tune in to approach control? On what frequency does he tune in to approach control? Who tells him to tune in to approach control?

8. Does the pilot request route clearance from ground control? With route clearance can he take off in bad weather? Do all large aeroplanes need route clearance? Who gives the pilot clearance? What clearance would a small aircraft need in good weather? What does the British Airlines pilot ask the air traffic controller?

9. At ramp two zero, is the pilot ready to taxi? Does he tune in to departure control at ramp two zero? What frequency does he tune in to at ramp two zero? What does he tell ground control he is ready to do while on the ramp?

10. When a radio message is finished, one says "Over". Does a pilot say "Over" at the end of a message? Do you say "Over" when you address me? Who says, "Over"? What does an air traffic controller say at the end of a message? The Pilot?

11. When a radio message is understood, one says, "Roger". Can a pilot say, "Roger" at the end of a message? Does a radio operator say, "Roger"? When does a first officer say, "Roger"? Do we say, "Roger", over the telephone? Who says "Roger"?

12. Does the ground controller relay route clearance to the pilot? Does he relay taxi clearance? To whom does he relay route clearance?

13. The pilot or first officer says, "Go ahead" to the ground controller. Does he say, "Go ahead" when he wants the ground controller to speak? Does he say, "Go ahead" when he wants to hear the message? What does he say when he wants the ground controller to give him a clearance?

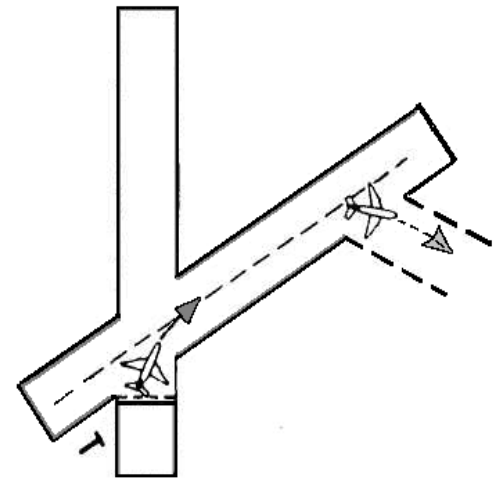
14. 15. Does the ground controller tell the pilot to maintain flight level 350? Does the ground controller tell him to maintain his climb? His cruising speed? What does he tell him to maintain? Is it safe to maintain a flight level? What flight level is the pilot told to maintain?

16. 17. The holding point is usually at the end of the taxiway. Does an aircraft wait at the holding point? Does a pilot contact aerodrome control at the holding point? Does he contact area control at the holding point? Approach control? Who does he contact at the holding point? Is the holding point clear of the runway? Is it on the runway? Is it clear of the taxiway? What is it clear of? Where is the holding point?

18. If runway 24 is 240 degrees, the heading is 240 degrees. Is the pilot instructed to continue runway heading? Is he told to continue runway heading after taxiing? When is he told to continue runway heading? What heading would he maintain on runway 24? 36? Walk to the door please. Where are you heading?

19. Does the ground controller tell the pilot to contact approach control? Does he tell the pilot to contact area control? Aerodrome control? Who does the ground controller tell the pilot to contact? At what stage of the journey does the pilot contact approach control? On what frequency does this pilot contact approach control? What does the controller actually say to the pilot? What does the pilot say to the ground controller when he taxis down the runway?

BA 624 has received clearance to line up and take off



WRITTEN EXERCISES

Instructions to students:

Write in your answers to the following questions giving complete sentences.

Example: What belts do passengers fasten at take-off?

Answer: At take-off, passengers fasten their seatbelts.

a.) When do passengers learn their emergency procedures?

b.) With what single word does one end a message on the radio?

c.) By tuning in to 124.6 frequency, what controller does the pilot on BA 624 contact?

d.) At what heading is runway 30?

e.) What view does a controller have from the tower?

f.) Why does an aircraft stand clear of the runway?

g.) With what two words do you ask a controller to give you route clearance?

h.) With what single word do you tell someone you have understood the message?

LESSON 2



AERONAUTICAL INFORMATION SERVICE (AIS)

Each country provides aeronautical information concerning its own territory. This is published in the Aeronautical Information Publication (1) (AIP) and in Notices to Airmen (2) (NOTAM).

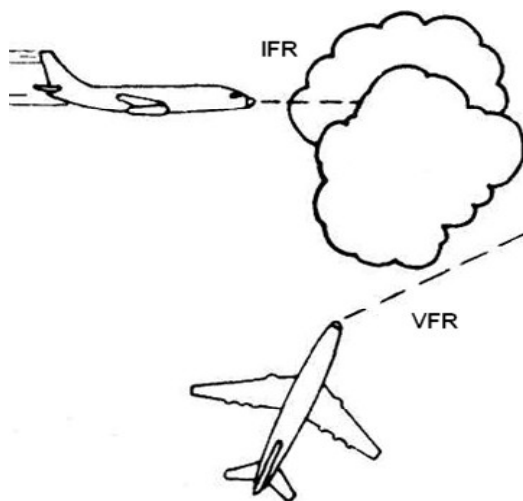
A pilot planning a flight will prepare his Flight Plan (3) from the information in these publications.

What sort of information does a pilot need to know when flying to another country or another aerodrome? For instance, he will want to know which airway (4) to follow.

Further information the pilot needs is, what sort of facilities (5) are available en route and at the aerodrome of his destination. Whether the aerodrome has suitable fuel available; paved runways; the length of these runways; the elevation of the aerodrome (6); what hangar space is available; the communication frequencies, and what meteorological information is available. This is some of the useful information that is in the AIP and NOTAM.

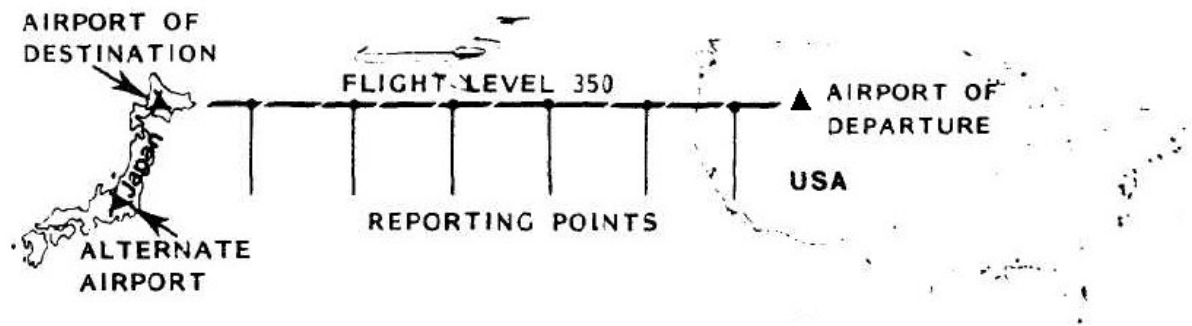
A pilot, therefore, who is flying from the United States to Japan presents himself to the Briefing Officer (7) at the aerodrome of departure. He fills out a Flight Plan giving the route he is to follow and the description of the route, the name of the aerodrome of his departure and that of his destination. Also the name of an alternate aerodrome (8). This is given in case the aerodrome of destination is closed when he gets there. He gives his cruising speed (9) in knots. Also he must indicate whether he will fly IFR (10) or VFR (11) or, a combination of both and moreover, he gives the flight level he wishes to maintain during the flight.

The pilot will have to report to Air Traffic Control at the various reporting points (12). With the information received from the pilot, air traffic control provides separation for his flight across the USA, the Pacific Ocean and Japan. The pilot must not stray (13) too far to the left or right of the centreline (14) of the airway. Of course a pilot must always maintain his approved altitude as many aeroplanes are flying at different levels in both directions and, unauthorized leaving of the airway or assigned flight level would be very dangerous.



VOCABULARY

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| 1. Aeronautical Information Publication | - Better known as AIP |
| 2. Notices to Airmen | - Better known as NOTAM |
| 3. Flight plan | - A document prepared by the pilot before his flight |
| 4. Airway | - A corridor generally comprising a width of 5 miles on each side of the centre line equipped with radio navigation aids. |
| 5. Facilities | - Services provided. These include buildings, such as hangars, or airport terminals, also communications, or fire fighting services etc. |
| 6. Elevation of aerodrome | - The height of the aerodrome above mean sea level. |
| 7. Briefing Officer | - The person who provides information before the beginning of the flight and receives the pilot's Flight Plan at the aerodrome of departure. |
| 8. Alternate aerodrome | - The aerodrome of second choice used when the aerodrome of destination is closed because of bad weather. |
| 9. Cruising speed | - This is the airspeed the aircraft maintains in straight and level flight. |
| 10. IFR | - Instrument Flight Rules. These rules must be used when visibility is poor but many flights are controlled in accordance with these rules even in good weather. |
| 11. VFR | - Visual Flight Rules. When visibility is good a pilot may fly in accordance with visual flight rules. |
| 12. Reporting points | - A specific geographical location in relation to which the position of an aircraft can be reported. |
| 13. To stray | - To deviate, to drift or go somewhere without intention. |
| 14. Centreline | - An imaginary line in the centre of an airway or a visual line in the centre of a runway. |

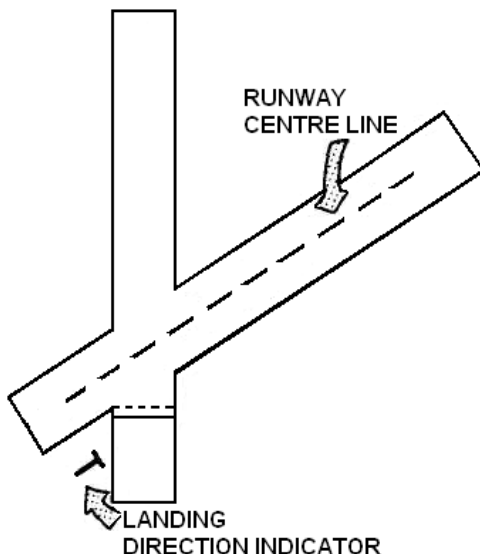
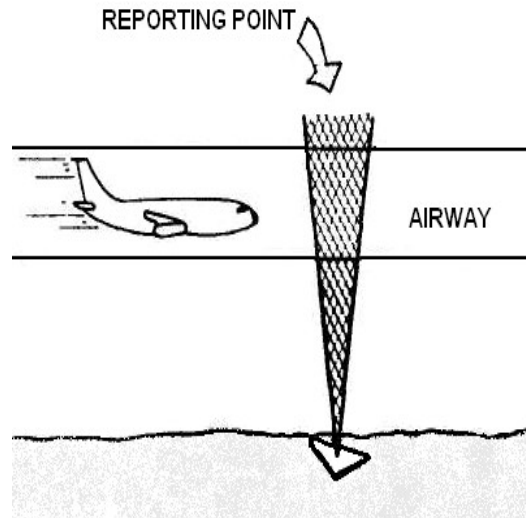


CONVERSATION

1. Is the AIP a publication? Is the AIP an international publication? Do countries provide aeronautical information for publication? Do countries provide aeronautical information concerning their own territories? In what publication is aeronautical information published? What does AIP stand for?
2. Is NOTAM a publication? Is NOTAM an international publication? Does NOTAM provide information to airmen? Does NOTAM provide information to you? To whom does NOTAM provide information?

Does the pilot need this information after his flight? When does the pilot need this information? What is NOTAM short for?

3. Does the pilot prepare his flight plan from the information in NOTAM? Does he prepare his flight plan before the flight? Does he prepare his flight plan after the flight? When does he prepare his flight plan?
4. Is an airway a corridor in the air? Is an airway a corridor on the ground? Where is an airway? What is an airway? Can you see an airway? Why can't you see an airway? What is the width of an airway? Are there many airways in the air?
5. Are there many facilities provided at most airports? Are there many facilities provided at some small airfields in the jungle? At what airports are there many facilities? Is the parking lot a facility? Tell me what are some of the facilities provided at big international airports?
6. The elevation of this aerodrome is 500 feet. The elevation of the aerodrome at Addis Ababa is 8,000 feet. What is the elevation of this aerodrome? What is the elevation of a house on the coast? What is the elevation of the aerodrome at Addis Ababa. How is the elevation measured?
7. Does the pilot give his flight plan to the briefing officer? Does he also give his flight plan to the stewardess? To whom does he give his flight plan? Is the briefing officer at the aerodrome of departure? Does the briefing officer provide information to the pilot? Does he provide information after the flight? When does the briefing officer provide information? To whom does the briefing officer provide information?
8. Does a pilot name an alternate aerodrome on his flight plan? Is an alternate aerodrome the closest aerodrome to his point of destination? Is an alternate aerodrome always in the same country of destination? What is an alternate aerodrome? Why does a pilot have to name an alternate aerodrome? On what document does he name an alternate aerodrome?



9. When a plane has climbed and reached flight level, can it then maintain cruising speed? When it is climbing, does it have cruising speed? Can it turn and bank at cruising speed? When does it have cruising speed?
10. Are pilots forced to fly IFR when the visibility is poor? Are they forced to fly IFR in fog? Do they have to fly IFR on a clear day? When would pilots have to fly IFR? What does IFR stand for? What does the pilot look at when he is flying IFR?
11. Are all pilots licensed to fly VFR? Do they fly VFR when the weather is good? Do they use instruments in VFR conditions? What are VFR conditions? What sort of licence do pilots need to fly VFR?
12. Are there reporting points in the USA? Are there reporting points across the Pacific Ocean? In Japan? What information does the pilot give the controller at the various reporting points?
13. 14. Is a pilot allowed to stray slightly from the centreline? Is he allowed to stray from the airway? Why is it dangerous for a pilot to stray outside the airway? Where is the centreline? Can we see the centreline in the airway? Can we see the centreline on the runway? Why can't we see the centreline in the airway?

WRITTEN EXERCISES

Instructions to students:

Write in your answers to the following questions:

a.) On the map, how many reporting points can you see across the Pacific Ocean?

b.) What is the flight level of this particular flight?

c.) Why is an alternate aerodrome necessary?

d.) On what document is the alternate aerodrome named?

e.) Who provides separation for flights flying in accordance with 1FR?

f.) What is the elevation of this aerodrome?

g.) Name three of the facility services at this aerodrome.

h.) Write in telecommunications terminology: 9600, 350, 24, and 7200.

i.) When a pilot tunes in to a radio frequency at departure to whom does he want to speak?

LESSON 3



A WEATHER FORECAST GIVES THE PILOT INFORMATION
ON WEATHER CONDITIONS WHICH MAY BE EXPECTED.

THE EFFECTS OF THE WEATHER ON AVIATION

Except perhaps for local or very short flights, a pilot, before taking off, obtains a weather forecast (1) giving him the weather conditions which are expected along the route of his flight and at his destination. Because weather conditions affect aircraft in flight, to a considerable extent, special aviation forecasts are provided by meteorologists (2) at weather offices all over the world.

The meteorologist, or forecaster, prepares a weather chart which shows the current weather conditions (3) over the whole country. The current weather chart is called a synoptic chart. (4) This synoptic chart shows the areas of low pressure, (5) the areas of high pressure, (6) where precipitation (7) is falling, and all other weather conditions across the country.

From this weather map, the forecaster can advise pilots of the weather conditions they can expect to encounter during their flights. A high pressure area, for instance usually means good weather while a low pressure area usually involves one or more fronts (8) producing clouds and precipitation over many hundreds of miles.

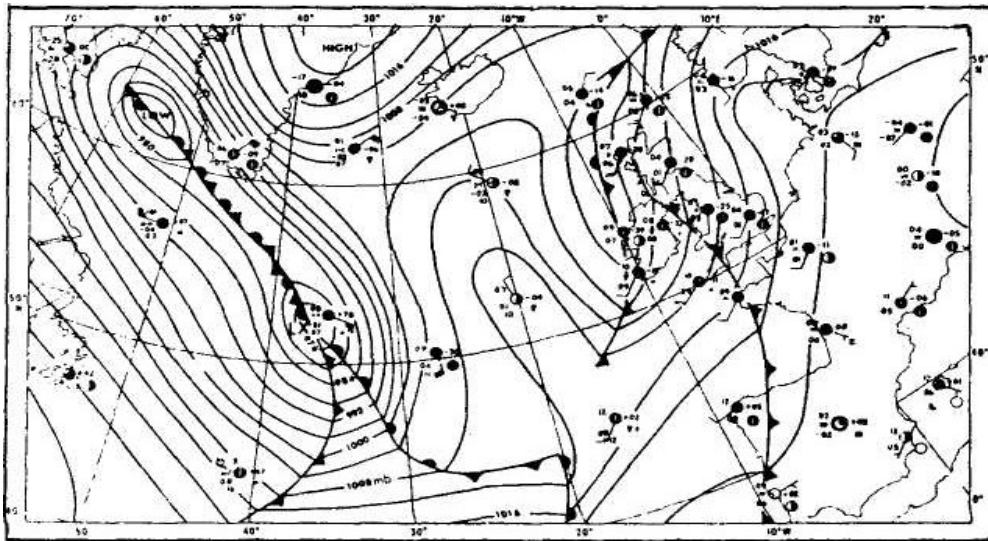
A pilot needs to know the wind direction and speed. A headwind (9) will obviously delay the arrival of flights and is to be avoided if at all possible. A tailwind (10) on the other hand, can be of great advantage as it increases the ground speed (11) and results in a reduction in fuel consumption. Winds vary with altitude, and also from one place to another, so information on winds is very important.

Pilots will pay particular attention to a low which lies en route, and the weather conditions associated with that low pressure area. The associated cold (12) or warm fronts (13) could involve clouds, thunderstorm, snow, rain, and turbulence. From his charts, the meteorologist can forecast where this weather will be at a certain time in the future, and with the help of these predictions, the pilot will decide which route to fly and when and he will know what weather conditions to expect. Should the forecast be very bad, for example, dense fog or poor visibility due to snow, the pilot may decide to postpone his flight. A pilot flying VFR would also cancel his flight because of low ceiling (14) or low overcast (15) conditions en route.

VOCABULARY

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|-------------------------------|--|
| 1. Weather forecast | - Weather prediction; future weather. |
| 2. Meteorologist | - The person qualified to predict a forecast of the weather. |
| 3. Current weather conditions | - The present weather conditions. |

- | | |
|---|--|
| 4. Synoptic chart | - Weather map showing the present weather conditions. |
| 5. Area of low pressure (low pressure area) | - An area as shown on the weather map when the pressure decreases towards the centre. |
| 6. Area of high pressure (high pressure area) | - An area as shown on the weather map when the pressure increases towards the centre. |
| 7. Precipitation | - Rain, snow, hail or any water that falls to the ground in liquid or solid form. |
| 8. Front | - An imaginary line on the ground which marks the division of the two air masses. |
| 9. Headwind | - The wind in the opposite direction to the travel of the aircraft. |
| 10. Tailwind | - The wind in the same direction to the travel of the aircraft. |
| 11. Ground speed | - The speed of an aircraft measured on the ground. This speed is affected by a headwind which would reduce it, or by a tailwind which would increase the ground speed. |
| 12. Cold front | - A cold front is formed when a moving cold air mass meets a stationary warm air mass. |
| 13. Warm front | - A warm front is formed when a moving warm air mass meets a stationary cold air mass. |
| 14. Ceiling | - The base of the clouds which cover more than 50% of the sky. |
| 15. Overcast | - The sky conditions when 90% or more is covered by clouds. |

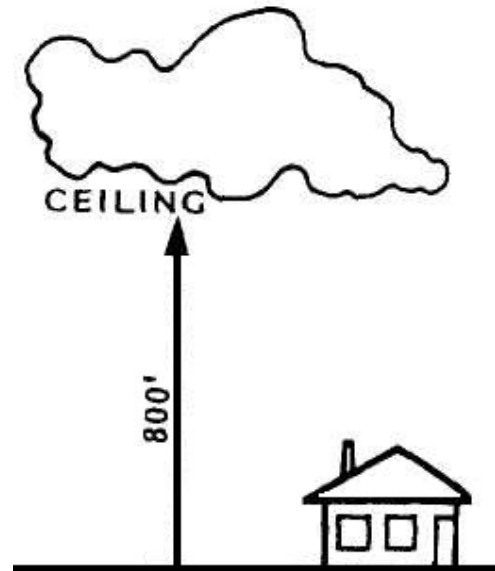


WEATHER MAP

CONVERSATION

1. I heard the weather forecast on the radio this morning. It said it was going to rain. Is it raining now? Does it always rain just because it is forecast? Is the weather forecast important to the pilot? To a sailor? To a motorist? To whom is the weather forecast important? What is the weather forecast for today? For tomorrow? When does a pilot need the weather forecast, before or after the flight?

2. A meteorologist is qualified to forecast the weather. Can a meteorologist forecast the weather? Can a meteorologist prepare a weather map? What can a meteorologist do? Can a farmer forecast the weather? A sailor? How does a farmer forecast the weather? A sailor? A meteorologist?
3. The current weather conditions in this area are good/poor/cloudy/overcast etc. Look out of the window and describe the current weather. Are you interested in the current weather? In tomorrow's weather? In yesterday's weather? Why are you only interested in the current weather and in tomorrow's weather? Why is the pilot more interested in the weather forecast than in the current weather?
4. Does a meteorologist prepare the synoptic chart? Does he prepare the navigational chart? What chart does a meteorologist prepare?
5. Does a low pressure area bring poor weather? Does a low pressure area also bring good weather? What sort of weather does a low pressure area bring? Can a low pressure area affect an area many hundreds of miles away? Can it bring clouds/ rain/snow/hail/other precipitation? Can a meteorologist show a low pressure area on his chart? Who is interested in a low pressure area? Why is a pilot interested in a low pressure area?
6. Does a high pressure area bring fair weather? Sunny skies? Clear days? Does a meteorologist show a high pressure area on a synoptic chart? How does he show a high or a low pressure area on his synoptic chart?
7. Is rain precipitation? Snow? Hail? Turbulence? Wind? Name three kinds of precipitation.
8. Is a headwind in front of the aircraft? Do pilots like headwinds? Does a plane consume more or less fuel with a headwind? Does a plane go faster or slower with a headwind? Does a plane arrive early or late with a headwind?
9. If headwinds are in front of a plane, are tailwinds in the rear? Are tailwinds good for fuel consumption? What winds do pilots prefer? Why do pilots prefer tailwinds?
10. If a pilot travels a distance of 100 miles in one hour is his ground speed 100 mph? If he travels a distance of 60 miles in one hour what is his ground speed? What type of wind affects his ground speed?
11. Is a front shown on a weather map? Does a forecaster show a front on a synoptic chart? Does he show a front on the synoptic chart to the pilot? To the navigator? To the radio maintenance technician? To whom does he show a front on the synoptic chart? How many types of fronts are there?
12. Do we find cold fronts on a weather map? On a navigation map? On what sort of maps do we find cold fronts? Does a meteorologist prepare a weather map showing the cold fronts? Does a pilot prepare a weather map showing the cold fronts? Who prepares a weather map showing the cold fronts?
13. Is a warm front also found on a weather map? Can pilots recognize a warm front on a weather map? Does everyone recognize a warm front on a weather map? Who recognizes a warm front on a weather map?
14. A low ceiling means that the clouds are low. Are the clouds low today? Do we have a low ceiling? Would the ceiling today trouble a pilot? Is 800 feet a low ceiling? 15,000 feet? What do you consider is a low ceiling of clouds? What effect can a low ceiling have on flying?
15. With a low ceiling are there many clouds in the sky? Are there more clouds in the sky when the sky is overcast? Can you see any blue sky when the sky is overcast? What colour is the sky when it is overcast?



WRITTEN EXERCISES

Instructions to students: The verb "to advise" and the noun "the advice" often cause some confusion. Similarly the verb "to affect" and the noun "the effect". In the following sentences choose the correct word.

a.) Forecasters can advise/predict pilots of the weather conditions.

b.) What advice/advice do pilots receive from forecasters?

c.) What affect/effect could a low pressure area have on an area hundreds of miles away?

d.) Does a low pressure area affect/effect an area hundreds of miles away?

e.) Snowstorms affect/effect aerodromes to the extent that all planes are grounded.

f.) What are the affects/effects of snowstorms on aerodromes?

g.) What affect/effect did the turbulence have on the passengers?

h.) Did the turbulence effect/affect the passengers?

i.) The engineer can advise/advice the pilot when fuel is low.

j.) The advice/advice the engineer gave the pilot was that his fuel was low.

k.) Will you advise/advice me of the precipitation in that area?

l.) I must receive advice/advice of the precipitation in that area.

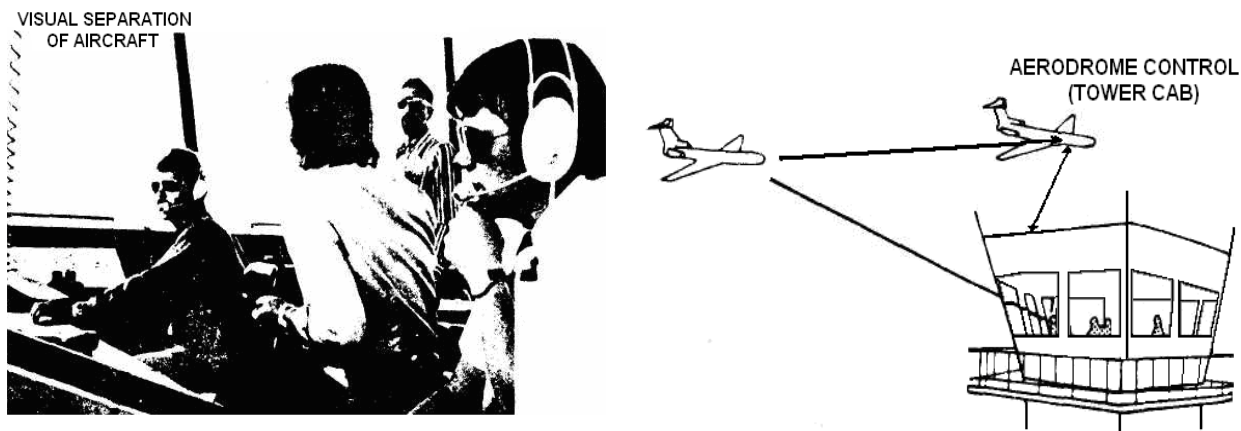
m.) The affects/effects of metal fatigue on aircraft can be very dangerous.

n.) Metal fatigue effects/affects aircraft and can be very dangerous.

LESSON 4

THE CONTROL TOWER

The control of air traffic on, or in the vicinity of an aerodrome, is provided by the control tower. In most cases the tower is located on top of the terminal building and, is restricted (1) to authorized personnel only. The aerodrome controller, who is located in the "tower cab", (2) provides separation between aircraft arriving and departing, and also controls aircraft and vehicles manoeuvring (3) on the manoeuvring area.



Approaching aircraft, particularly those operating under VFR, generally contact the aerodrome controller while still several miles away. The aerodrome controller informs the pilot of the runway in use, (4) the wind direction and speed, the altimeter setting, as well as information on other traffic in the area.

Approaching the aerodrome, the pilot joins the traffic circuit (5) on the downwind leg. (6) The aerodrome controller then advises the pilot of his landing sequence (7) giving such instructions as, "You are number two to land following the Cessna on final approach", (8) or, "You are number three following the 707 on base leg". (9)

The tower's "tools" include a number of radio transmitters and receivers used to communicate with pilots, and a signalling lamp (10) to send control signals to any aircraft not radio equipped or with a radio that is not properly working. The tower's radio equipment may be used with microphone (11) and loudspeakers (12) or with a headset. (13) There is a wind speed indicator(14) and a wind direction indicator (15) at each controller console. (16) A further instrument is the altimeter setting indicator. (17) It is from this instrument that the controller advises the pilot of the current altimeter setting which the pilot sets on his aircraft altimeter.

At some busy airports an Automatic Terminal Information Service (ATIS) is available to pilots of departing and arriving aircraft. This service consists of a continuous radio broadcast on a special frequency of recorded and periodically updated no control information. Included is information regarding ceiling, visibility, wind direction and speed, altimeter settings, and runway in use. Where ATIS is not available, the tower provides this information.

VOCABULARY

- | | | |
|-----------------|---|--|
| 1. Restricted | - | On the ground an area limited to authorized persons. In the air, an (area) airspace within which flying is restricted in accordance with certain conditions. |
| 2. Tower cab | - | The upper part of the control tower wherein aerodrome controllers work. |
| 3. To manoeuvre | - | To move around. A manoeuvring area is that part of an aerodrome to be used for take off and landing of aircraft and for the surface movement of aircraft associated with take off and landing, excluding aprons. |

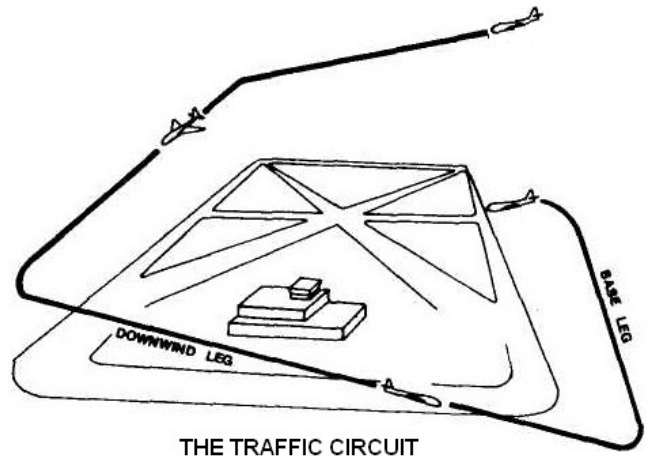
4. Runway in use - The runway used at any given moment for take-offs and landings.
5. Traffic circuit - A pattern which aircraft follow for landing consisting normally of downwind leg, base leg, final approach. (See below.)
6. Downwind leg - The downwind part of the traffic circuit. Downwind is the direction measured relative to the way the wind is blowing? Downwind means the direction in which the wind is moving.
7. Landing sequence - One's turn. Coming after or next to land. Whether number one, number two, number three, etc.
8. Final approach - The last part of the traffic circuit prior to landing. Also called "final".
9. Base leg - The crosswind part of the traffic circuit between downwind and final.
10. Signalling lamp - A light showing a red, white, or green signal for controlling aircraft without radio. Also called a light gun or an Aldis lamp.
11. Microphone - The instrument one speaks into when using the radio or sound system.
12. Loudspeaker - An apparatus that converts electrical impulses into sound so that it is heard at some distance.
13. Headset - An apparatus that is worn on the head by pilots and controllers. It has a microphone and an earpiece (speaker).
14. Wind speed indicator - An instrument measuring the speed of the wind.
15. Wind direction indicator - An instrument showing the direction of the wind indicator
16. Console - The furniture at the controller's position holding instruments, microphone, radar display etc.
17. Altimeter setting indicator - An instrument at the control tower which indicates the altimeter setting to be transmitted to aircraft.

CONVERSATION



1. Can controllers go to the control tower? Can aircraft mechanics go to the control tower? Why can't the mechanic go to the control tower? Because it is a restricted area. Can aircraft fly low over a city? Why not? Can you name any other restricted area?
2. Do aerodrome controllers work in the tower cab? Do area controllers work in the tower cab? Who works in the tower cab? Where is the tower cab? What can controllers see from the tower cab?
3. When a pilot taxis his aeroplane, does he manoeuvre it? Does he manoeuvre it on the taxiway? On the apron? Does he go back and forth when he manoeuvres his aircraft? Where does he manoeuvre his aeroplane? Can pilots manoeuvre their planes easily in lots of open space? Can they manoeuvre their planes in crowded places? When can they manoeuvre their planes easily? Name the areas where aircraft manoeuvre.
4. Do planes use the runway in use? Are all the runways at an aerodrome in use at the same time? How many runways are in use at the same time? In what direction generally is the runway in use facing? Why does the runway in use generally face the wind?
5. Do all aeroplanes join the traffic circuit? Do aeroplanes descend in the traffic circuit? Do they wait to land in the traffic circuit? What do aeroplanes do in the traffic circuit?

6. Is the downwind leg part of a plane's descent? Is the pilot flying into the wind on the downwind leg? Who tells him to join traffic at the downwind leg? Can a controller tell a pilot to follow a plane on the downwind leg? What is a pilot doing on the downwind leg?
7. When a controller advises a pilot of his landing sequence, is he telling him he is next to land? Last to land? Cannot land? What is the controller saying to the pilot when he advises him of his landing sequence?
8. At final approach can the pilot see the runway? At final approach can the controller see the aircraft? Is the aircraft close to the ground at final approach? At what part of the traffic circuit is final approach? What clearance does the controller give the pilot at final approach?
9. Is base leg also a part of the traffic circuit? Is a plane landing on base leg? Is it also taking-off on base leg? What is an aeroplane doing on base leg? Is base leg before final approach? Higher, lower than final approach? Downwind leg? Where is base leg? What can a pilot see on base leg? How many turns does he have to make on base leg? Is he far from the runway on base leg? Where is the plane while on base leg?
10. Is a signalling lamp a light for signalling aircraft? Is it also called a light gun? An Aldis lamp? Can one fire a light gun? What does a controller do with a signalling lamp? Where does he use a signalling lamp? What colour signals does signalling lamp have? What does a green light mean? Red? White? Why does a controller need a signalling lamp? Give me some other names for a signalling lamp.
11. Does a controller need a microphone? Does he need a microphone if he uses a signalling lamp? When does a controller need a microphone? Where is the microphone? Where is the pilot's microphone? Have you ever used a microphone?
12. Are there loudspeakers in the tower cab? In the aeroplane? Can many people listen to messages from the loudspeaker? Can only one person listen to messages from the loudspeaker? How many people can listen to messages from the loud-speaker? Who can hear the loud-speaker in the control tower? In the passenger cabin? Can one talk into a loudspeaker?
13. Does a pilot wear a headset on the flight deck? In the cabin? Walking through the terminal? Where does a pilot wear a headset? The controller? Why does he wear a headset? Does a headset have a microphone? Is the microphone fixed to the headset? What else is on he headset?
14. 15. Does a wind speed indicator show the speed of the wind? Velocity? Direction? What does a wind speed indicator do? What does a wind direction indicator show? Are these indicators to be found in the control tower? On an aeroplane instrument panel? Where are they to be found? How does a pilot find out the direction of the wind? Name another wind direction indicator on the aerodrome.
16. Is a console a piece of furniture? Is a console in front of the controller? The pilot? Where is the console? What instruments are on the console?
17. Is an altimeter setting indicator an instrument? Does a pilot have an altimeter setting indicator? What does an altimeter setting indicator show? How does a pilot make sure that his altimeter setting is correct?



WRITTEN EXERCISE

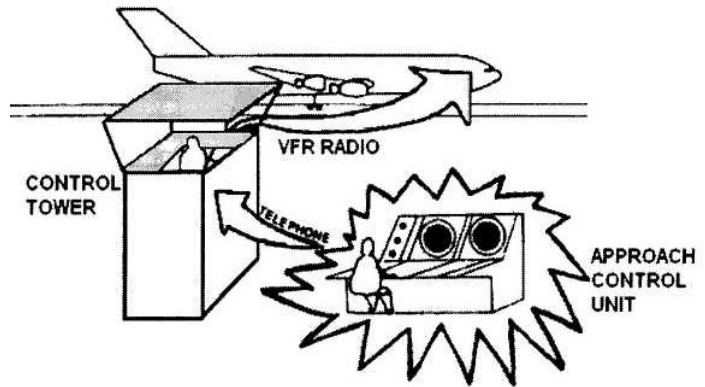
Instructions to the students: Write in your answers on a separate sheet of paper, using full sentences.

- a) Name the three parts of the traffic circuit in order of sequence.
- b) Describe the use of the signalling lamp.
- c) Why is ATIS of the help to the controller?
- d) Name some of the visual references you use to determine the direction of the wind.
- e) Describe some of the equipment in the tower cab, as well as their uses.

LESSON 5

APPROACH CONTROL

Approach control can be situated at the aerodrome or at some other site. Normally, if there are two aerodromes in a vicinity, there will be one approach control unit to cover them both. Generally speaking, the approach controller's main task is to provide separation to departing and arriving aircraft operating under instrument flight rules. The approach controller generally takes over the control of an arriving aircraft from the airways controller and eventually hands over that aircraft to the control tower. For departing aircraft the procedure is reversed. A considerable degree of (1) co-ordination is therefore required between the various units in air traffic services.



CO-ORDINATOR BETWEEN UNITS IN ATS

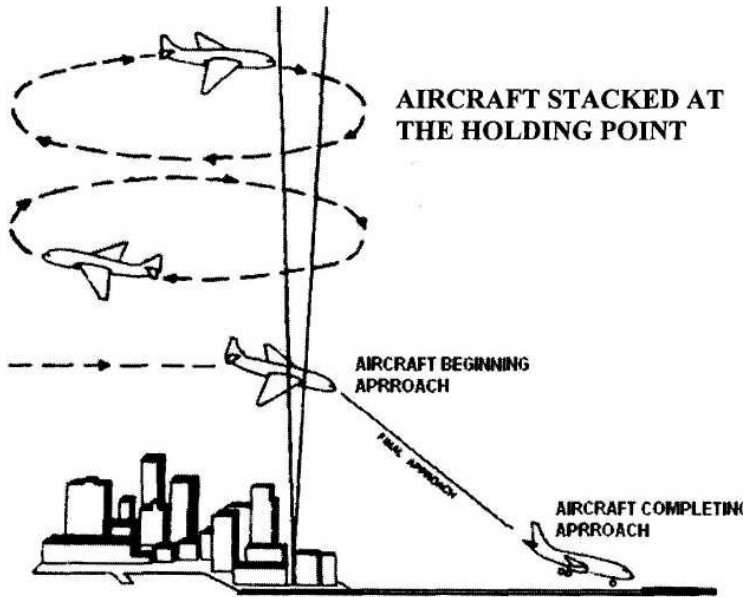
The approach control unit has radio equipment similar to that in the control tower. This enables the controller to communicate with aircraft on several different radio frequencies. Most of the frequencies are in the (2) VHF range with some in the (3) UHF.

Aircraft operate under two different types of rules: IFR and VFR. If (4) Visual Meteorological Conditions prevail, flight under Visual Flight Rules (VFR) Coordination between unit in ATS is permissible. In (5) Instrument Meteorological Conditions, Instrument Flight Rules (IFR) apply.

When airports are very busy the approach controller may have (6) to stack the aircraft at different levels so that each may descend in turn towards the airport. In such cases the approach controller usually gives aircraft their (7) expected approach time. Because of their high fuel consumption at low levels, jet aircraft and, particularly (8) supersonic aircraft, prefer to remain at higher altitudes before starting their descent.

VOCABULARY

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| 1. To co-ordinate: | - To pass all information from one controller to another. |
| 2. VHF: | - Very High Frequency. |
| 3. UHF: | - Ultra High Frequency. |
| 4. Visual Meteorological Conditions: | - Known as VMC. Clear weather conditions equal to or better than, specified minima which permit a flight to be conducted in accordance with Visual Flight Rules (VFR). |
| 5. Instrument Meteorological Conditions: | - Known as IMC. Poor weather conditions, less than the minima specified for VMC. |
| 6. To stack: | - To place one on top of the other. |
| 7. Expected Approach Time (EAT): | - The time at which ATC expects that an arriving aircraft following a delay, will leave the holding point to complete its approach for landing. |
| 8. Supersonic aircraft: | - Aircraft flying over Mach 1 (faster than the speed of sound). |



CONVERSATION

When many people are working at similar tasks, they must co-ordinate (1) their work. Do military pilots flying in formation coordinate with each other? Is it necessary for an architect to co-ordinate with an engineer? Approach controller with aerodrome controller? Area controller with aerodrome controller? Who does an area controller co-ordinate with? Approach controller? Military pilots flying in formation? Tell me what you think, would happen if military pilots did not co-ordinate with each other.

Is VHF (2) very high frequency? Can we transmit messages on VHF? To whom does the controller speak on VHF? What does VHF mean?

Is UHF (3) another radio frequency? Is UHF ultra high frequency? Can we tune-in to UHF? Can we tune-in to UHF on an ordinary radio? On what type of radio can we tune-in to UHF?

Is the weather clear under Visual Meteorological Conditions? Do pilots like to fly under Visual Meteorological Conditions? (4) Can he see the airport under Visual Meteorological Conditions? Can the airport be closed-in under Visual Meteorological Conditions? What sort of weather is Visual Meteorological Conditions? What is the abbreviation of Visual Meteorological Conditions? What rules do pilots follow under VMC?

Is the weather cloudy under Instrument Meteorological Conditions? (5) Is it overcast? Is visibility good under Instrument Meteorological Conditions? What sort of weather is it under Instrument Meteorological Conditions? What is the visibility under IMC? What conditions do we have today, IMC or VMC?

Do controllers stack (6) flights at busy airports? Do they stack flights at quiet airports? At what airports do controllers stack flights? Why do they stack flights? Who stacks flights? How do they stack flights? Where do they stack flights?

Does a pilot receive his expected approach time (7) from the controller? Does he receive his expected approach when he is at the holding position? On base leg? On the downwind leg? Where is the pilot when he receives his expected approach timer? Who gives the pilot his expected approach time? What is the pilot doing when he receives his expected approach time? What is the abbreviation of expected approach time?

The Concord is a supersonic aircraft. (8) Does a supersonic fly faster than the speed of sound? Does the supersonic fly very high? Does the supersonic consume a lot of fuel? Is the Boeing 747 supersonic? What plane is supersonic? What other supersonic aircraft do you know? Is there any special radiotelephony phraseology for supersonic aircraft?

WRITTEN EXERCISE

Instructions to students:

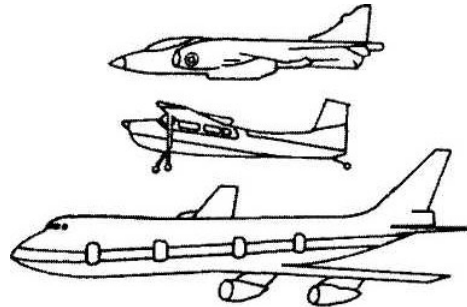
What is the difference between the job of an aerodrome controller and an approach controller? Describe in your own words.

LESSON 6

A CONVERSATION BETWEEN A FLIGHT DISPATCHER AND A PILOT

A pilot can give his flight plan to an ATS Unit or he can hand it to his flight dispatcher. In fact, that is what the First Officer of UN 739 was doing when the conversation between them turned to the pilot's destination. Flight Dispatcher: "So you are heading for O'Hare Aerodrome? Isn't it the busiest aerodrome in the world?"

Pilot: "It certainly seems that way when you get into that traffic circuit. I have flown in there many times and the traffic is always stacked."



THREE TYPES OF AIRCRAFT

- | | |
|--------------------------|--|
| Flight Dispatcher | - "What <u>type of aircraft</u> (1) are they for the most part?" |
| Pilot | - "Every type from jets to the smallest Cessna. However, although it is an <u>international airport</u> , (2) its traffic is mainly <u>domestic flights</u> ." (3) |
| Flight Dispatcher | - "I suppose London is the busiest airport for international traffic." |
| Pilot | - "Yes, of course, it has all the international traffic crossing Europe. I was stacked there on a <u>holding point</u> (4) for half an hour one morning waiting to get my approach clearance. It was in turbulent conditions too, and the passengers were not very happy. By the time we were on downwind leg we were almost an hour behind schedule." |
| Flight Dispatcher | - "Imagine how much fuel you must have consumed. Come to think of the Concord takes-off and lands at Heathrow. A supersonic like that must be stacked at a very high level when it has to hold. Have you ever flown the Concord?" |
| Pilot | - "No, I have only flown <u>subsonic</u> ." (5) |
| Flight Dispatcher | - "Before they use up all the fuel in the world I want to fly on Concord. Then when we have <u>reverted</u> (6) back to the horse and shall sit in the village square and tell all the kids about how flew on the Concord." |
| Pilot | - "They will probably say, "Poor old gaffer - he never did, you know |

VOCABULARY

- | | |
|--------------------------|--|
| 1. Type of aircraft | - The make; the model of aircraft e.g. DC8; Boeing 747. Also category of aircraft e.g. jet aircraft, twin engine aircraft, etc. |
| 2. International airport | - Airport of entry and departure for international air traffic. |
| 3. Domestic flights | - Flights which operate in the country in which the aircraft is registered. |
| 4. Holding point | - A location over which an aircraft in flight is maintained in accordance with ATC clearance. In the vicinity of airports aircraft are stacked over the holding point. |
| 5. Subsonic | - Jet aircraft flying below Mach 1. |
| 6. To revert back | - To stop advancing and go back. |

CONVERSATION

1. Is a Cessna a small type of aircraft? Is it a twin engine type aircraft? A jet? What type of aircraft is a Cessna? A Beechcraft? A Concord? A Boeing 747?
2. Does an international airport receive flights from other countries? Does it receive gliders? Balloons? What type of aircraft does an international airport receive? Where do planes come from that operate in and out of international airports? Name some of the well known International airports?
3. Do domestic flights also use international airports? Do domestic flights follow the same traffic circuit as international flights? Do domestic flights have their own aerodrome? Is military aviation domestic? Are planes from other countries domestic? What planes are domestic? Domestic flights follow the same safety standards as international flights?
4. Is a holding point situated above an aerodrome? Are flights stacked at a holding point? Why are they stacked at a holding point? Who gives them clearance to leave the holding point? What is another name for a holding point? Where are flights generally going from a holding point, up or down?
5. Is a Boeing 707 a subsonic aircraft? Are military planes usually subsonic? What type of plane is a B-707? Military jet? Concord? etc.
6. Do you like cars? Radios? Motorbikes? Would you like to revert back to the days before cars and radios? Why wouldn't you like to revert back to the 19th century? What would we lose if we reverted back to the 19th century?

WRITTEN EXERCISES

Instructions to students: Using the following vocabulary or expressions, invent appropriate air traffic controllers' instructions to pilots.

Example: "Hold short of, you say something like: "Hold short of runway two fower ". Write on a separate sheet of paper. You may use more than one of the following words or expressions in the same sentence.

- | | |
|-----------------------------|---------------------------------------|
| a. cleared to | g. roger |
| b. taxi to | h. go ahead |
| c. tune in to | i. via flight plan route |
| d. contact approach control | j. hold short of |
| e. heading | k. flight level 290 |
| f. over | l. the current weather conditions are |

LESSON 7

AREA CONTROL SERVICE

The function of area control is to provide air traffic control service for controlled flights except for those parts of flight where the ATC service is provided by approach control or by aerodrome control. Its area of control usually includes many thousands of square miles of airspace. With radar, controllers can position aircraft and achieve the desired safe and expeditious flow of traffic. (However, area control can be provided without radar). Some centers have computers (1) which automate many of the routine functions (2, 3) of the controller. In order to maintain a controller's workload (4) at a level within his capability, the centre's airspace is divided into sectors. (5)

It is essential that the controller know the position and future plan of every aircraft within his sector. To accomplish this, flight progress boards (6) are used on which are placed flight progress strips (7) which contain all the pertinent current flight data (8) such as the following:

Aircraft identification (9)

Type of aircraft

Point of departure

Destination

Flight planned route

Filed true airspeed (10)

The controller's and pilot's estimated time over the current reporting point

Actual time over reporting point or fix

Flight level in hundreds of feet

Clearance information

Where a centre does not have the requisite computer, flight progress strips are manually (11) prepared and handled.

Approach control facilities use radar and flight progress strips in the same manner as area control centres.

A controller has to decide which separation standard he will apply to aircraft in flight. If he applies lateral separation (12) he must maintain aircraft on different routes or in different geographical areas.

In applying longitudinal separation (13) the controller maintains an interval between aircraft. Longitudinal separation is established by requiring aircraft to depart at a specified (14) time; to arrive over a reporting point at a specified time; or to hold over a reporting point until a specified time. A 15-minute, time-spacing interval between two 600 mph jet aircraft means that they are separated longitudinally by 150 miles.

Vertical separation (15) is obtained by assigning different flight levels to aircraft, in other words, they are separated by a specified vertical distance.

VOCABULARY

- | | | |
|----------------------------|---|--|
| 1. Computer | - | Electronic equipment for processing information mathematically and logically. |
| 2. Routine | - | Procedure that does not change from day to day. |
| 3. Functions | - | Work; employment. |
| 4. Workload | - | Quantity of work. |
| 5. Sector | - | A portion of airspace assigned to a controller. |
| 6. Flight progress board | - | A board designed and used for the display of flight information. |
| 7. Flight progress strips | - | Usually in the form of a piece of paper measuring about 2 x 15 centimeters containing flight progress information and kept on the flight progress board used in control units. |
| 8. Current flight data | - | Up-to-date flight information. |
| 9. Aircraft identification | - | A group of characters given to each aircraft to distinguish it from others. |
| 10. Filed true airspeed | - | The airspeed entered on the flight plan. |
| 11. Manually | - | Handled by hand. |
| 12. Lateral separation | - | Separation by maintaining aircraft at different routes or in different |

- | | |
|-----------------------------|---|
| | geographical areas. |
| 13. Longitudinal separation | - Separation of aircraft by maintaining a time interval between them. |
| 14. To specify | - To say definitely. |
| 15. Vertical separation | - Separation of aircraft by assignment of different levels. |

CONVERSATION

1. Do we use computers in aviation?

Do computers help process flights? Are flights processed faster because of computers? Do computers only process flights? What information can computers process? Why are computers necessary in aviation? Why do controllers like computers?

2. I get up every morning of the week, wash, have breakfast and come here. This is my routine every morning of the week. If I go for a walk in the country, is this my routine? What is my routine?

Is filing a flight plan the routine work of the pilot? Flight Progress Strips in an area control centre

Is checking the checklist the routine work of the pilot?

Why is checking the checklist, routine?

Why is filing a flight plan, routine?

What would happen if it were not routine?

3. The functions of a controller are mainly to separate flights. Are his functions very responsible? Are his functions very difficult? Busy? Routine? Why are some of his functions routine? Do some centres have computers for the routine functions of the controller? Do all centres have computers for the routine functions of the controller? What centres have computers for the routine functions of the controllers? Do computers automate these functions?

4. Is the workload of a controller at a busy aerodrome very heavy? Is the workload of a flight attendant sometimes very heavy? On an empty flight is a flight attendant's workload very heavy? On what sort of flight is a flight attendant's workload heavy? Do you have a heavy workload in this lesson? Will you have a heavy workload when you study air traffic control? When does a pilot have a heavy workload? An air traffic controller? You?

5. Is a control area divided into sectors? Is a control area divided into sectors when the workload is heavy? Is a control area always divided into sectors? When is a control area divided into sectors? Is a controller responsible for his own sector? Are there many flights in each sector? Does a controller separate the flights in his own sector? Why does a controller handle only one sector?

6. Does a controller have a flight progress board? Is the flight progress board in front of him? Is the flight progress board sometimes behind him? Where is the flight progress board?

7. Does a controller enter information on a flight progress strip? Does he enter the information on the flights in his sector? What information does he enter on his flight progress strips? Approximately what size is a flight progress strip?

8. On a flight progress strip a controller enters only current flight data. Does he enter only current data? Does he enter yesterday's data? What data does he enter on a flight progress strip? Who enters the data? Does he enter data on the flights in his sector? Tell me some of the data a controller enters on flight progress strips.

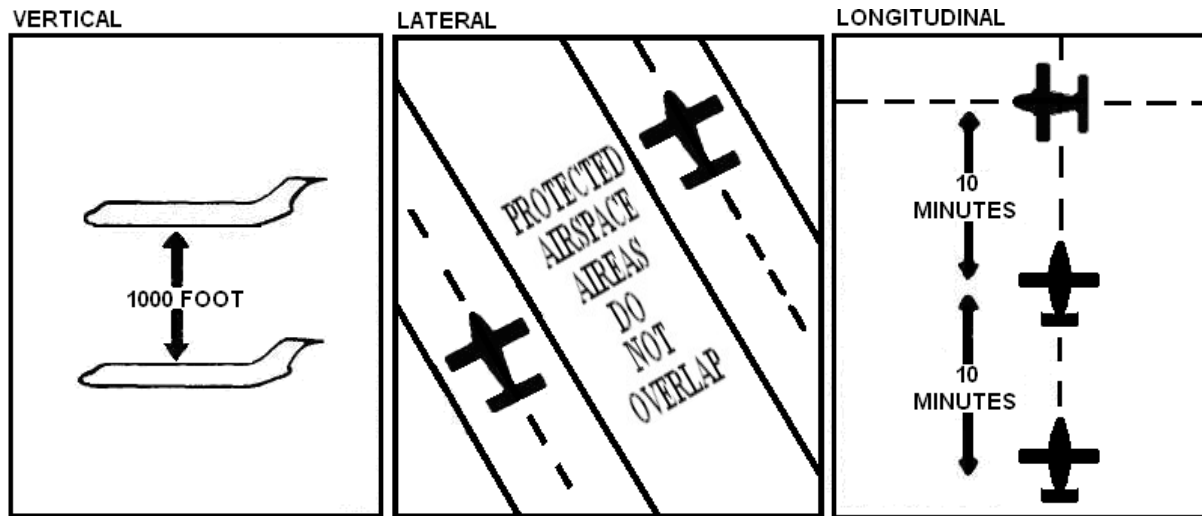
9. Is an aircraft identification a group of characters used to identify the aircraft? Is an aircraft identification a number? A group of numbers? A group of letters? Is an aircraft identification a name? A colour? Is an aircraft identification a number of a flight? What is an aircraft identification? What is BA 624? Oscar Foxtrot Charlie?

10. On the flight plan the pilot has entered the true airspeed of the flight. Does a pilot enter the true airspeed of the flight on the flight plan? Does a controller enter the true airspeed on the flight progress strips? Where does the pilot enter the true airspeed? Where does the controller enter the true airspeed? Is the true airspeed part of the data on the flight progress strip?

11. Does a pilot control his plane manually? Does he always fly from A to B manually? When does he control the plane manually? Does a driver always control a car manually? Does a controller always prepare his flight progress strips manually? When does a controller prepare flight progress strips manually?

12. In lateral separation is there a safe distance between flights? In lateral separation is there a safe distance on the right? On the left? Behind? Below? In lateral separation where is there a safe distance? Who provides lateral separation? Under what weather conditions does a pilot provide lateral separation? A controller?

13. In longitudinal separation is there a safe distance in front? Behind? Below? Above? Where is there a safe distance in longitudinal separation?



14. We start our lessons each day at a specified time of 0900 hours. Do we finish our lessons at a specified time? What is the specified time for starting our lessons today? Tomorrow? What is the specified time for finishing our lessons today? Tomorrow? Do you go to the cinema at a specified time? Do you have your meals at a specified time? Do you always go to bed at a specified time? Does longitudinal separation require pilots to depart at a specified time? Report at a reporting point at a specified time?

15. Is vertical separation when aircraft fly at different flight levels? Is vertical separation important to all aviation? Do planes fly in both directions in vertical separation? Does vertical separation involve different flight levels? Who controls vertical separation of flights? Why?

WRITTEN EXERCISES

Instructions to students: Choose the appropriate meaning or meanings. Mark the square or squares accordingly as there might be more than one that would apply.

a) An aircraft is identified by

- a.) its colour. ☐
- b.) its type. ☐
- c.) a designator and its flight number. ☐

b.) When aircraft depart at specified times they are separated

- a.) one behind the other. ☐
- b.) vertically. ☐
- c.) by time. ☐
- d.) longitudinally. ☐

c.) VHF is

- a.) a meteorological terminology. ☐
- b.) a navigational aid. ☐
- c.) a radio frequency. ☐
- d.) an instrument on the flight deck. ☐

d) A windsock is

- a.) another word for downwind leg. ☐
- b.) a wind direction indicator. ☐
- c.) visible to pilots. ☐
- d.) a constant movement. ☐

d) An Aldis lamp is

- a.) in the control tower. ☐
- b.) a signalling lamp. ☐
- c.) in an area control centre. ☐
- d.) a light gun. ☐

e) Overcast conditions generally prevail at

- a.) the sea. ☐
- b.) the tropics. ☐
- c.) the mountains. ☐

f) It is easier to manoeuvre with a

- a.) jeep. ☐
- b.) fire engine. ☐
- c.) bus. ☐
- d.) B-747. ☐

g) The elevation of the aerodrome at Addis Ababa is

- a.) 35,000 feet. ☐
- b.) high. ☐
- c.) at mean sea level. ☐
- d.) low. ☐

h) An alternate aerodrome is filed on a

- a.) flight plan. ☐
- b.) licence. ☐
- c.) flight progress strip. ☐
- d.) rating. ☐

i) An aircraft can fly on automatic

- a.) at take-off. ☐
- b.) on landing. ☐
- c.) when en-route. ☐
- d.) when on straight and level. ☐

LESSON 8

RADIO NAVIGATION AIDS - VOR/DME

Apart from taking their bearings (1) from the stars, how do pilots navigate? The answer, of course, is in the use of radio navigation aids. (2) There are a variety of different types of radio navigation aids but here we shall discuss VOR (3) and DME (4) only.



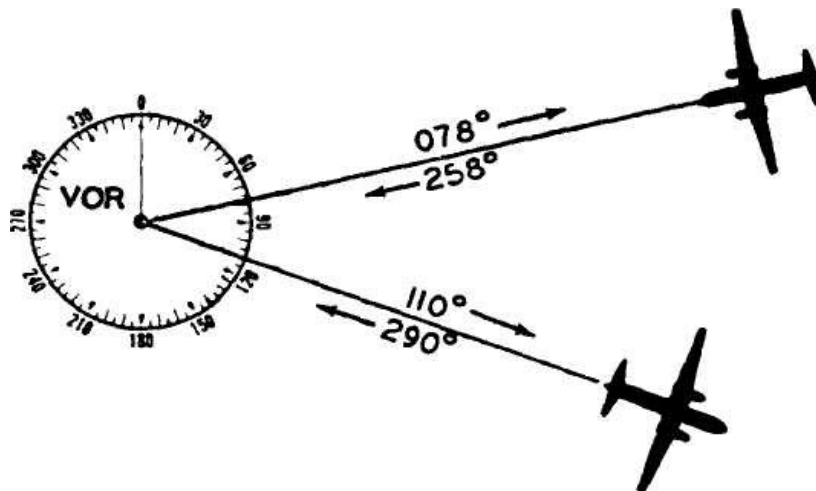
VOR and DME are often located at the same site. They operate on VHF and UHF respectively and, as a consequence, are not affected by static (5) or other interferences. (6) The maximum range of VOR is about 200 nautical miles. (7) By flying VOR the pilot ensures he is flying directly to the station. Also by measuring his radials (8) from more than one VOR station, a pilot can check his position.

The function of DME, which is short for Distance Measuring Equipment, is, as its title describes, to measure distance.

The DME measures, electronically, the time it takes for a signal, transmitted from an aircraft interrogator (9), to reach the ground base station transponder (10), and return.

This elapsed time is converted to miles and appears on a digital indicator on the flight deck. The indicator actually seems to rapidly count the number of miles between the aircraft and the station giving the pilot a continuous digital reading (11) of how far he is from, or to, a station.

With the many VOR/DME stations along his route, a pilot can make good his desired track; is constantly aware of his distance to or from a DME station; or, by using two VOR radials, establish his exact position.



THE OMNI RANGE GIVES THE PILOT
HIS EXACT BEARING AT ALL TIMES,
TO OR FROM THE STATION

VOCABULARY

1. Bearing - The angle between a direction and a reference direction as determined at the place of the observer. The reference direction is generally the North.
2. Radio navigational aids - Also known as navaids. Electronic equipment for navigation.
3. VOR - Very High Frequency Omni directional Range.
4. DME - Distance Measuring Equipment.
5. Static - In this case it is noise on the radio caused by atmospherics.
6. Interference - Intrusion of another station making communication difficult or impossible.
7. Nautical miles - A measurement of distance used primarily in navigation. It is equal to 6080 feet or 1852 meters and is one minute of latitude at the equator.
8. Radials - Airways or routes to and from a VOR station.
9. Interrogator - Electronic equipment in the aircraft for navigation by DME.
10. Transponder - A ground based receiver/transmitter which receives a radio signal from an aircraft and transmits it back to the aircraft.
11. Reading - In this case it is the figure or number (of miles) shown on the DME indicator on the aircraft instrument panel.

CONVERSATION

By taking his position from the stars, a navigator can find his bearings. (1) Can a navigator find his bearings from the stars? Can a pilot find his bearings from the stars? Can you find your bearings from the stars? Who can find his bearings from the stars? Who can find his bearings from the use of navaids? (2) Does a map help you find your bearings when you are lost? Name different ways in which a pilot could find his bearings.

Is VOR (3) a navaid? Is VOR located on the ground? Is VOR also on the flight deck? Where is VOR? Does a pilot use VOR in order to navigate? Can he get his bearings with VOR? Why does a pilot use VOR?

Is DME (4) a navaid? Is DME short for Distance Measuring Equipment? Can a pilot navigate with DME? Can he find his bearings with DME? Is DME a piece of electronic equipment? Is DME on very high frequency? UHF? In air traffic control? Where is DME? What does DME stand for?

Is static (5) a noise on the radio? Can you hear a radio programme if there is much static? Does static interfere (6) with the radio programme? Is static a pleasant noise? Do you like listening to static? Why don't you like listening to static? Does static interfere with VOR? DME? Why doesn't it interfere with VOR/DME? Can interference come from another station? Do you like listening to the radio with interference? Static? How do you like listening to the radio?

Are nautical miles (7) used in navigation? Do navigators measure distances in nautical miles? Do pilots measure distances in nautical miles? You? How do you measure distances? Pilots? Navigators?

Are radials (8) lines on a pilot's map? Do radials extend from VOR stations? Are radials on a road map? On what sort of map are radials? From where do radials extend?

Is an interrogator (9) a piece of electronic equipment? Is an interrogator on the flight deck? At an air traffic control centre? Where is an interrogator?

Is a transponder (10) also a piece of electronic equipment? On the ground? At a DME station? At a railway station? At what station is a transponder? What sort of equipment is a transponder?

The time by my watch is 0300 hours. What is the time by your watch? Is the reading (11) of your watch the same as mine? Is the reading of the pilot's altimeter the same as the air traffic controller's? What is the reading of this thermostat? Have you checked the reading on the thermostat? Can the pilot check the reading on his altimeter? Can the flight attendant check the reading on the altimeter? Who checks the reading on the altimeter?

WRITTEN EXERCISES

Instructions to students: Answer the questions in your own words using full sentences. Write on a separate sheet of paper.

When a navigator uses a compass, the stars, VOR, a map, etc. what is he trying to find?

- a.) Why does a pilot need both VOR and DME?
- b.) What does the DME indicate to the pilot?
- c.) In order to get a correct reading on his altimeter does a pilot check with air traffic control before he lands?
- d.) What radio frequencies are VOR and DME operated on? And are they affected by static?
- e.) Does an aircraft have to be fitted with instruments in order to receive radio signals?

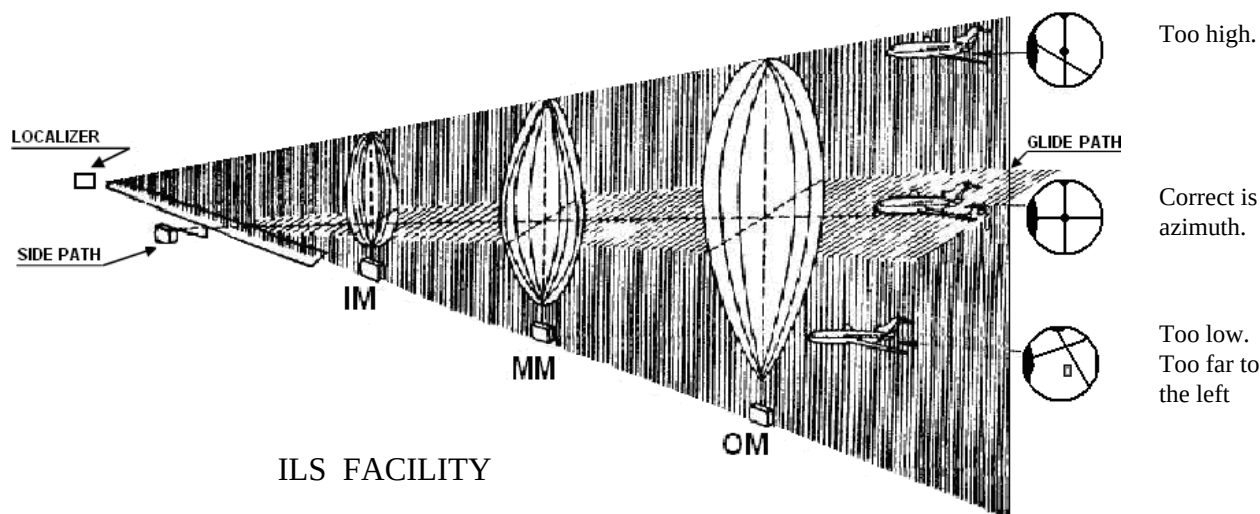
LESSON 9

RADIO NAVIGATION AIDS AIDS TO FINAL APPROACH AND LANDING ILS

By using VOR and DME a pilot knows he is on the correct course and he also knows his distance to the VOR/DME station. With this equipment the aircraft will be brought directly over the VOR/DME station from which point an approach can usually be made.

In poor weather conditions, however, a more precise (1) landing aid is used. The ILS (2) is the ICAO (3) approved international standard electronic landing aid and is installed at nearly all major aerodromes.

The ILS functions with equipment in the aircraft and on the ground. On the aerodrome a localizer (4) transmits a narrow radio beam (5) and another transmitter provides a glide path (6) at a fixed angle of approach. On the aircraft instrument panel an ILS indicator (7) shows the aircraft's position in relation to the centre line of the runway and to the glide path. The ILS can, therefore, guide the aircraft along the proper approach path down to a point where the pilot must be able to see the ground and be able to continue his approach to land. If he cannot see the ground at this point, he must decide to overshoot (8), go around and try to land again.



In addition to the ILS localizer and glide path, an ILS installation also comprises two or three fan markers (9): one called the outer marker (10) is situated from four to six miles from the threshold (11) of the runway; a second called the middle marker (12) is situated approximately half a mile from the threshold of the runway, and the third, called the inner marker (13) (installed only when required) is about 1,000 feet from the threshold of the runway. These fan markers alert the pilot of his passing over these markers by causing a light on the aircraft instrument panel to flash on and off while he is over each marker.

VOCABULARY

- | | |
|-------------------------|--|
| 1. Precise | - Exact; correct; accurate. |
| 2. ILS | - Instrument Landing System. |
| 3. ICAO | - The International Civil Aviation Organization. |
| 4. Localizer | - Part of the ground based ILS equipment. |
| 5. Beam | - A ray of light or of electronic radiation. |
| 6. Glide path equipment | - Part of the ground based ILS equipment. |

- | | |
|-------------------|---|
| 7. ILS indicator | - The ILS instrument on the pilot's instrument panel. |
| 8. To overshoot | - To cease descending (usually on final approach) and begin climbing.
- The terms to pull-up or go around are also used. |
| 9. Fan marker | - Electronic equipment transmitting radio signals in the shape of a fan. |
| 10. Outer marker | - One of the fan markers. |
| 11. Threshold | - The beginning of that portion of the runway usable for landing. |
| 12. Middle marker | - - do - |
| 13. Inner marker | - - do - |

CONVERSATION

What time is it? Is it precisely 0300 hours? Is your watch precise? **(1)** Is Big Ben precise because it is on GMT? Is this clock precise? What clock is precise? Is ILS a more precise form of landing aid than VOR? Does a pilot landing through low clouds need a very precise landing aid?

Is ILS **(2)** a landing aid? Is ILS short for Instrument Landing System? Is ILS short for Instrument Meteorological Conditions? What is ILS short for? What landing aid do pilots use?

Is ICAO **(3)** an international organization? Is this manual printed by ICAO? What is ICAO short for? Who printed this book?

Is the localizer **(4)** a part of the Instrument Landing System? Is the localizer on the ground? Is the localizer also in the aircraft? Where is the localizer? Does the localizer send signals to the aircraft? Does the localizer transmit a beam? A vertical beam? A horizontal beam? What sort of beam does the localizer transmit? At what sort of aerodromes are the localizers to be found? Is the localizer beam narrow or wide?

When I switch on a flash light is there a beam **(5)** of light? Is there a beam of light when I switch on the light in this room? When is there a beam of light? Does electricity radiate a beam? Does electricity radiate a beam of light? Can a transmitter radiate a beam? A beam of light? What sort of beam does a transmitter radiate? A flash light? A localizer?

Is the glide path **(6)** near the runway? Is the glide path located near touchdown? Where is the glide path located? Why is the glide path necessary in the Instrument Landing System? Does the glide path transmit a beam? In what direction does the glide path transmit a beam? Is the glide path beam narrow? Wide? What sort of beam does the glide path transmit?

Is the ILS indicator **(7)** on the pilot's instrument panel? Is the ILS indicator also on the ground? Where is the ILS indicator? Are there two needles or one on the ILS indicator?

Does a pilot sometimes decide to overshoot? **(8)** When a pilot overshoots does he pull back on the control column? What happens to the aircraft when the pilot overshoots? Does it take less power or more power to overshoot? Less fuel or more fuel? At what stage of approach would a pilot decide to overshoot? Give me other words for overshoot.

Is a fan marker **(9)** at the aerodrome? At the reporting point? Where is a fan marker? Is the fan marker a navaid? Is a fan marker an electronic piece of equipment? Fixed equipment? Mobile equipment? What sort of equipment is a fan marker? What does a fan marker transmit?

Is an outer marker **(10)** a fan marker? Is an outer marker located miles out from the runway threshold? How many miles out is the outer marker?

Is the runway threshold **(11)** on the runway? In the middle of the runway? At the end of the runway? Where is the runway threshold? What does a plane do at the runway threshold, take-off or land?

Is the middle marker **(12)** in the middle of the runway? At the beginning? On the threshold? Where is the middle marker?

Is the inner marker **(13)** the last marker on the glide path? Is the inner marker also on the aerodrome? Where is the inner marker at the aerodrome? How does the pilot know he is flying over the inner marker? What other markers are at the aerodrome? What service do they give to the pilot?

Instructions to students: Describe in your own words an aircraft approach at an airport under IMC due to fog, and how the ILS helps the pilot in the approach to the runway. Write on a separate sheet of paper.

LESSON 10

CONVERSATIONS

Between a Pilot and a Flight Attendant

- Pilot:** Well! Miss Jones, are you looking forward to our flight to Tokyo?
- Flight attendant:** With three hundred and twenty passengers on board, I am expecting a very busy flight. I should imagine I shall be very tired by the time we reach Tokyo. What is our ETA? (1)
- Pilot:** 0800 hours. We are going to be a little late – full load (2) and headwinds. Our ground speed will be less than 450 knots (3). I can see the first of your passengers coming aboard. I shall see you dinner time.
- Flight attendant:** Before you go, Captain, what is our flight level and do you think the flight will be smooth?
- Pilot:** We shall be cruising at 39,000 feet – we couldn't get 35,000 because of traffic. The forecaster promised no turbulence so it should be smooth.

Between a Meteorologist and an Air Traffic Controller

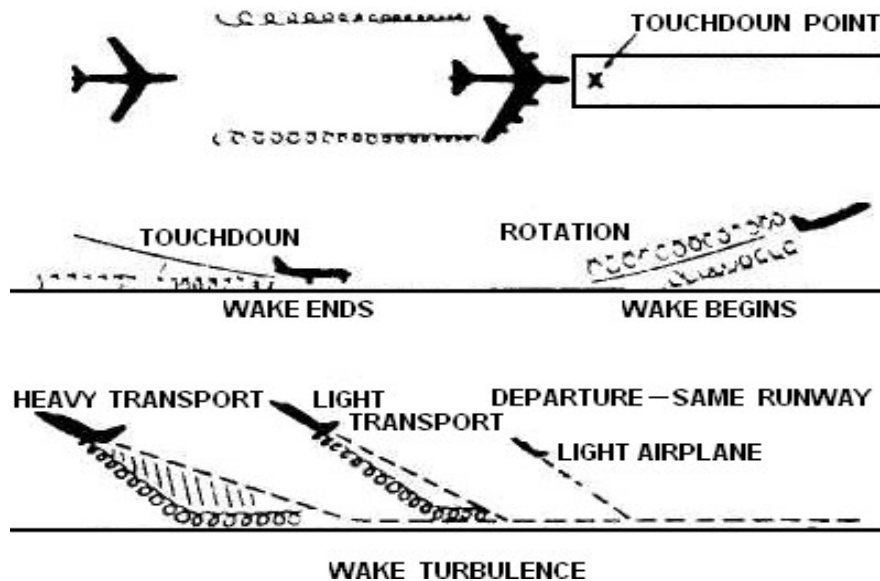
- Controller:** Hello, Steve. I notice some clouds moving in from the West. Are we going to have a change in the weather?
- Meteorologist:** The forecast we gave you this morning is still valid except that the warm front is moving in towards us a little faster than forecasted. The barometer (4) is dropping rapidly and I believe the rain will probably begin by 0200 hours.
- Controller:** So I guess we are in for a wet night?
- Meteorologist:** It may get worse – the spread between (5) the temperature and dew point (6) is two degrees so we may get fog before the front reaches us.
- Controller:** OK, I'll be prepared for poor visibility. Thanks, Steve, and good-night.

A BAROMETER



Between two Air Traffic Controllers

- Tower controller:** What a busy time we had in the tower tonight. I suppose it was not any easier in the centre?
- Centre controller:** It was hctic. (7) The cold front was the cause of the trouble; we had to vector (8) most of the aircraft around thunderstorms.
- Tower controller:** We had trouble with wake turbulence. (9) It was worse than usual on runway 24. We had to increase the separation between landing aircraft, particularly for the smaller planes. Plus that, there were a lot of local flights. (10)
- Centre controller:** Many of those local flights were probably the training flights from the local flying school.
- Tower controller:** Yes, they are always very active when the weather is above VFR limits. I wonder what the forecast is for tomorrow.
- Centre controller:** It is supposed to be good.
- Tower controller:** Well, I guess we are in for another busy day. Good-night, Dick.
- Centre controller:** Good-night, Tom, see you tomorrow.



Between a Pilot and a Passenger who is visiting the flight deck

- Pax:** I have always wanted to see what a cockpit looked like on a DC-8. I have a private pilot licence and fly a Cessna 172.
- Pilot:** Welcome aboard. This cockpit looks more complicated than it actually is.
It is probably easier to fly than the Cessna 172. We are just about to make a, procedure turn (11) in the holding pattern. (12) We shall keep turning until we intercept (13) the inbound (14) radial to the fix. (15)
- Pax:** What nav aids are you homing on? (16)
- Pilot:** VOR. We shall continue to fly the holding pattern until given further clearance by the controller. We have already been given an expected approach time – about three minutes from now in fact.
- Pax:** In that case I had better get back to my seat. Thank you for letting me visit you.

VOCABULARY

1. ETA - Estimated time of arrival.
2. Full load - Filled to utmost capacity.
3. Knots - A measurement of speed.
4. Barometer - An instrument measuring atmospheric pressure.
5. The spread between - The difference between.
6. Dew point - The temperature at which dew forms.
7. Hectic - Extremely busy.
8. To vector - To guide a pilot in flight.
9. Wake turbulence - Turbulent air behind a large aircraft which can turn over a small aircraft following too closely.
10. Local flights - Flights that take off and land at the same aerodrome.

- | | | |
|---------------------|---|--|
| 11. Procedure turn | - | A manoeuvre in which a turn is made e.g. a turn made in a holding pattern. |
| 12. Holding pattern | - | A predetermined circuit which keeps an aircraft within a specified airspace. |
| 13. To intercept | - | To reach/arrive at an airway or radial. |
| 14. Inbound | - | Towards an airport, a navaid etc. (the opposite is outbound). |
| 15. Fix | - | Any navaid from which an aircraft can determine its exact position. |
| 16. To home-on | - | Heading towards a radio station or navaids. |

CONVERSATION

Is ETA an abbreviation for estimated time of arrival? Is ETA the exact time of arrival? What is ETA? Who wants/needs to know the ETA of a flight? Why would you want to know the ETA of a flight? When a plane has a full load (1) is it heavy? Can a plane with a full load travel fast? Does a plane arrive early with a full load? What can make a full load on an aeroplane? Do all planes have a full load? How does an airline know the plane has a full load?

Do planes measure their speed in knots? (2) Is a plane travelling at 500 knots going fast? What type of plane travels at 500 knots a 747 or a Cessna? If a barometer measures the atmospheric pressure, is it useful in aviation? Does a meteorologist use a barometer? Does a mechanic use a barometer? Who uses a barometer? What does a barometer measure?

The difference between the temperature and the dew point (3) is called the spread. (4) Could a narrow spread between the temperature and the dew point bring fog? Poor visibility? Does a pilot need to know the spread between the temperature and the dew point? The meteorologist? The navigator? Who needs to know the spread between the temperature and the dew point?

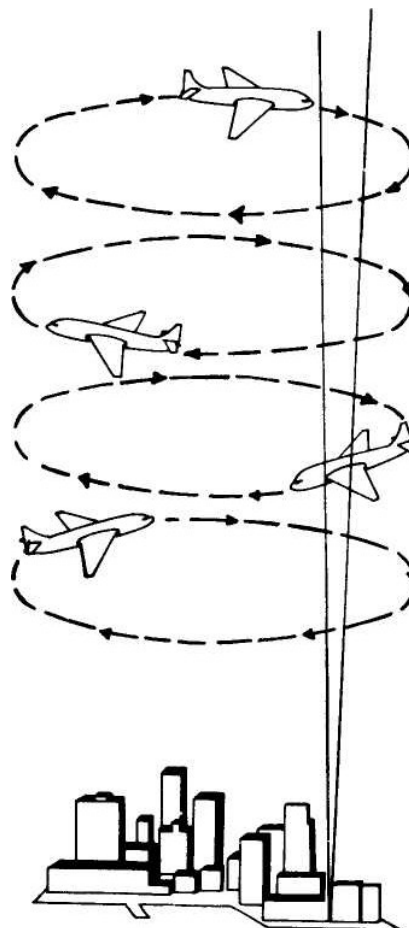
Do controllers sometimes have a hectic (5) time with the traffic? Do they always have a hectic time? When do they have a hectic time?

A controller vectors (6) a pilot when he says turn left 20°; turn right 25° etc. Who vectors a pilot? Does he use radar to vector a pilot? Can he vector a pilot from the tower cab? What controller vectors a pilot? Why does a pilot need to be vectored?

Is wake turbulence (7) dangerous? Does wake turbulence form behind large aircraft? At take off? At landing? Where is wake turbulence to be found? What type of aircraft leave wake turbulence behind them?

Do local flights (8) take off and land at the same airport? Does a flying school have local flights? Does a local flight pilot need a flight plan? Does a local flight travel far? Does a local flight need much fuel? From where does a local flight take off and land?

Does a pilot use his instruments to make a procedure turn? (9) Does he use his rudder to make a procedure turn? His undercarriage? What does a pilot use to make a procedure turn?



THE HOLDING PATTERN

Is the holding pattern (10) above an airport? Can many planes take their turn to land in a holding pattern? Is it always necessary to hold in the holding pattern? When is it necessary to hold in the holding pattern? What turn does a pilot make in the holding pattern?

Does a pilot crossing the Indian Ocean intercept (11) an airway? Crossing Africa? Does a controller tell a pilot to intercept an airway at 35,000 feet? What does a controller actually say to a pilot when he wants him to intercept an airway at 35,000 feet? 39,000 feet?

Does an inbound (12) aircraft head towards an airport? Does an inbound aircraft head towards a VOR? DME? ILS? What does an inbound aircraft do? Who sees an inbound aircraft from the tower? What traffic pattern does an inbound aircraft join?

Where does an inbound aircraft join the holding pattern? In what direction is an inbound aircraft heading? What is the opposite of inbound? Is a fix another word for navaid? Is VOR a fix? Tell me the other navaids that are a fix. Does a radial lead to a fix?

When a pilot follows a VOR radial, does he home-on (13) the VOR? The DME? The ILS? What navaid does he home-on on a VOR radial? A DME radial?

WRITTEN EXERCISES

Instructions to students: Using the following vocabulary or expressions, invent appropriate air traffic controllers' instructions communications for pilots.

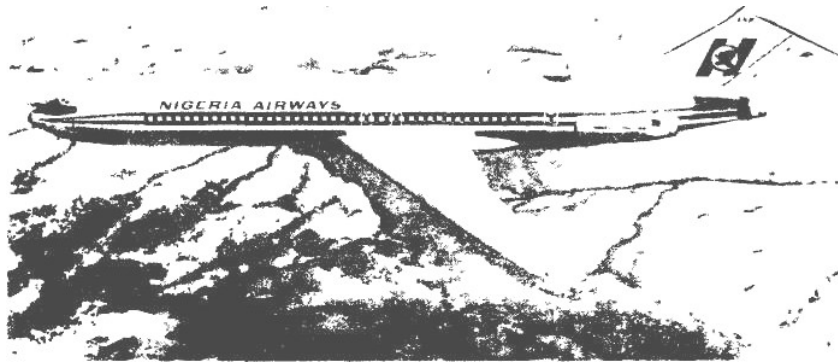
Example: "to intercept", you could write the following: "Turn right ten degrees to intercept 'J' ". Or another example could be "passengers". You could write "Please advise how many passengers are on board". Write on a separate sheet of paper. You may use more than one of the following in any exercise.

- | | |
|---------------------------|--------------------|
| a. a procedure turn | f. holding pattern |
| b. expected approach time | g. downwind leg |
| c. barometer | h. hold short of |
| d. dew point | i. cruising speed |
| e. wake turbulence | j. maintain |

LESSON 11

READING EXERCISE

Note to Instructor. The objective of this lesson is twofold, to practice pronunciation, and to enhance the students' understanding of words and terms introduced in previous lessons. For greater effectiveness, it is suggested that students take turns in reading aloud the parts of the pilot, the controller, and the narrator.



NIGERIAN AIRWAYS FLIGHT 267 EN-ROUTE FROM LAGOS TO ATHENS

APPROACH AND LANDING

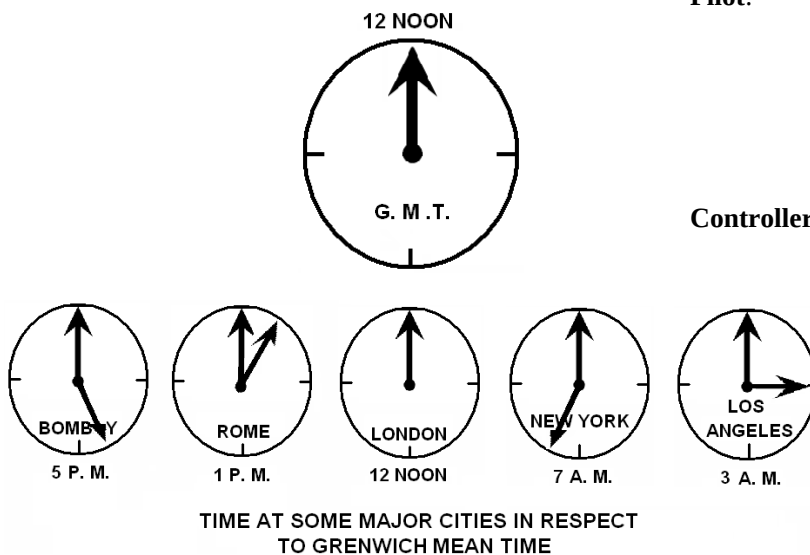
Narrator: Nigerian Airways Flight 267 departed Lagos, Nigeria at 1245Z (Greenwich Mean Time) enroute to Athinai (Athens), Greece on an IFR Flight Plan. Weather conditions on departure were VMC and no turbulence was expected during the flight. However, during his pre-flight weather briefing, the pilot had been informed by the forecaster, to expect reduced visibility on arrival at his destination. Flight 267 is now cruising at 35,000 ft. and is just passing over the Benina VOR Station. The pilot contacts the Tripoli area control centre.

Pilot:

Tripoli Control, Nigerian two six seven, over Benina at one six two fower, flight level tree fife zero estimating Paleothora at one seven zero one, Athinai at one seven one eight. Request descent clearance.

Controller:

Nigerian two six seven, Tripoli Control, roger, cleared to the Paleothora intersection, descend to and maintain flight level tree one zero. Hold over Paleothora at flight level tree one zero until further advised. Expect onward clearance at one seven one zero. Report leaving flight level tree fife zero. Present weather Athinai, ceiling indefinite, sky obscured, visibility five kilometres.



Pilot: Tripoli Control, Nigerian 267. Cleared to the Paleothora intersection, descend to flight level tree one zero. Hold over Paleothora intersection, at flight level tree one zero. Expect onward clearance at one seven one zero. We are leaving flight level tree fife zero now.

Controller: Nigerian two six seven, roger.

Narrator: Flight 267 can now begin descending from its cruising altitude of 35000 ft. to its present assigned altitude of 31,000 ft. but cannot go past the Paleothora intersection, which is about 260 nautical miles northeast of Benina, until it receives further clearance from Air Traffic Control. The Controller in the Tripoli ACC expects that further clearance will be available by 1710 Z. This means that Flight 267 may expect a ten minute delay in the holding pattern at Paleothora. This delay is due to restricted visibility at Athinai which is causing delays to aircraft arriving and departing. Flight 267 has now reached 31,000 ft. and reports to ACC as follows.

Pilot: Tripoli Control, Nigerian two six seven flight level tree one zero.

Controller: Roger. Nigerian two six seven, maintain flight level tree one zero. Contact Athinai Control on one two fower decimal fower.

Pilot: Nigerian two six seven, changing to Athinai Control on one two fower decimal fower.

Narrator: A considerable amount of coordination, between Air Traffic Services units, had been effected prior to the delivery of the above clearances to Flight 267. This coordination took place not only between the Tripoli and Athinai area control centres, but also with the Malta ACC. This coordination with Malta ACC was required because, while flying on the VOR airway between Benina and Paleothora, Flight 267 will fly over a small segment of the Malta FIR. The clearances given to Flight 267 are the result of coordination between these three Air Traffic Control units. It is now time for Flight 267 to contact Athinai ACC.

Pilot: Athinai Control Nigerian two six seven - flight level tree one zero estimating Paleothora at one seven zero one.

Controller: Nigerian two six seven, Athinai Control, roger. Now cleared to Athinai VOR, no delay expected. Descend to and maintain flight level one niner zero. Report over Paleothora.

Pilot: Athinai Control. Nigerian two six seven. Cleared to Athinai VOR, no delay expected, descend to and maintain flight level one niner zero.

Narrator: Good news for Flight 267! The flight will now be able to proceed directly to Athinai without executing a holding procedure over Paleothora. Having reached that intersection, the captain contacts Athinai Control once more.

Pilot: Athinai Control Nigerian two six seven. Over Paleothora on the hour descending to flight level one niner zero.

Controller: Nigerian two six seven. Athinai Control, roger. Contact Athinai Approach Control on one two two decimal one.

Narrator: Flight 267 will soon be entering the Athinai Approach Control area which extends to about fifty miles from the Athinai airport. In the Approach Control unit, controllers will be using modern automated radar systems which display, on the radar scopes, the aircraft's position, and the aircraft's identification for those aircraft equipped with transponders. With this information, the controller can provide the necessary separation between aircraft and maintain an expeditious flow of traffic which will minimize delays. Flight 267 as directed by Athinai now contacts the Approach Control unit.

Pilot: Athinai Approach, Nigerian 267, on one two two decimal one.

Controller: Nigerian 267, Athinai Approach.

Pilot: Nigerian 267 is over Paleothora on the hour, descending to flight level one niner zero, estimating Athinai at 17.

Controller: Nigerian 267, roger, now cleared to the Athinai VOR, via Milos One arrival, descend to and maintain flight level one tree zero, report passing flight level one fife zero.

Pilot: Nigerian 267 cleared to the Athinai VOR via Milos One arrival, descend to and maintain flight level one tree zero, report passing flight level one fife zero.

Narrator: Flight 267 is now cleared to the Athens VOR "via Milos One arrival". This is one of the approved arrival procedures which are published by the Aeronautical Information Service (AIS) and contained in the Aeronautical Information Publication (AIP) of Greece. Controllers and pilots are familiar with all these procedures for the airports they serve. Flight 267 is now at 15,000 feet, still descending, and now informs the approach control unit as requested.

Pilot: Athinai Approach, Nigerian two six seven, passing flight level one fife zero.

Controller: Nigerian 267, roger. Weather: sky obscured, visibility five kilometers in haze, wind two niner zero fife knots. Descend to and maintain altitude 5000 feet transition level seven zero, QNH one zero one fower.

Pilot: Athinai Control Nigerian two six seven, copied OK, descending to altitude 5000 feet QNH one zero one fower.

Narrator: The captain of Flight 267 now has all the information he requires for his landing at Athinai: the altimeter setting (1014) which he has now set on his altimeter; the runway in use (runway 33 right); the arrival procedure he will use (Milos One); the wind speed and direction (290 degrees at 5 knots). Although the visibility is only five kilometers, the weather will not pose any problems except that a visual approach will not be possible. The flight is now reaching 5000 feet and the pilot reports to ATC.

Pilot: Athinai Approach Control, Nigerian 267, now reaching five thousand feet.

Controller: Nigerian 267, Athinai Approach, cleared for ILS approach runway three three right. Contact Athinai Tower on one one eight decimal one.

Pilot: Athinai Approach, Nigerian, cleared ILS runway three three right, changing to Tower one one eight decimal one.

Pilot: Athinai Tower, Nigerian 267 on one one eight decimal one.

Controller: Nigerian 267, Athinai Tower, runway three three right, wind two nine five, six knots, QNH one zero one four, continue ILS approach, report outer marker.

Pilot: Athinai Tower, Nigerian 267, QNH one zero one four runway three three right, will report outer marker.

Narrator: A few minutes later.

Pilot: Athinai Tower, Nigerian 267, outer marker.

Controller: Nigerian 267, Athinai Tower, cleared to land.

Pilot: Athinai Tower, Nigerian 267, cleared to land.

Narrator: When the aircraft has landed, the tower controller will give Flight 267 his exact landing time and hand over the control to the ground controller who will provide the captain with taxi instructions to the apron where the flight from Lagos to Athens will come to an end.

LESSON 12



THE PILOT ALIGNS HIMSELF WITH THE RUNWAY

VISUAL AIDS FOR NAVIGATION MARKINGS AND LIGHTS

Additional visual aids to navigation consist of markings on the aerodromes. These markings comprise single lines or rows of lines which, for the pilot, are very important for holding positions, runway thresholds, the runway centre lines, the sides of the runways, etc. etc...

However, at night or during poor visibility by day, lights are required. For lights to be effective they must be of adequate intensity. (1) At certain aerodromes the controller can vary the intensity of some of the lights so that they can be reduced to avoid momentarily blinding the pilot and strong enough so that he can see them in thick weather.

The first light a pilot sees on approach is generally the aerodrome beacon. (2) It may rotate (3) and can be seen at a great distance distinguishable by its coloured flashes (4) alternating with white, or white only. There might be an identification beacon (5) which shows green flashes of light which identify the aerodrome in morse code. (6) Red lights, the usual danger signal, warn pilots of the obstacles such as hangers and other high buildings, telephone poles, etc. Runway edge lights (7) identify the runway and approach lights assist the pilot to align (8) himself with the runway.

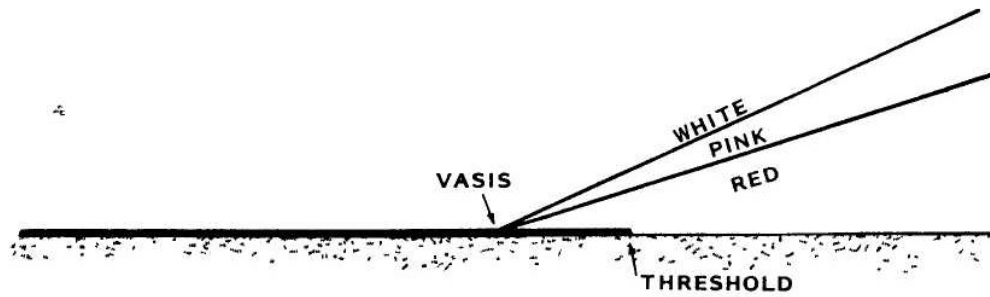
Lights may also be used to provide a glide path similar to what an ILS provides electronically. The Visual Approach Slope Indicator System, better known as VASIS (9), is a beam of light having a white colour in its upper part and a red colour in its lower part. A pilot of an aeroplane during an approach will:

- a) when above the approach slope, see the lights to be white in colour;
- b) when on the approach slope, see the lights to be pink in colour; and
- c) when below the approach slope, see the lights to be red in colour.

By reference to VASIS, combined with ILS, the pilot can bring an aircraft down safely almost to touchdown by day or night.

After landing, he follows the blue taxi lights along the taxiway to the apron and the service areas.

At the service area a marshaller (10) with illuminated wands (11,12), directs the aircraft with signals to its proper position for unloading and, finally, signals pilot to cut his engines. (13)



A SIMPLE VISUAL APPROACH INDICATOR SYSTEM

VOCABULARY

- | | |
|--------------------------|---|
| 1. Intensity | - Level of brightness. |
| 2. Aerodrome beacon | - Aerodrome light used to indicate the location of an aerodrome from the air. |
| 3. To rotate: | - To revolve; move round axis or centre. |
| 4. Flashes | - Lights that go on and off. |
| 5. Identification beacon | - A light identifying an aerodrome. |
| 6. Morse code | - Signal by long or short flashes when lights are used and by dots and dashes when telegraph is used. |
| 7. Runway edge | - The sides of the runway. |
| 8. To align | - To bring into line. |
| 9. VASIS | - Visual Approach Slope Indicator System. A glide path formed by lights. |
| 10. Marshaller | - The signaller who directs the pilots on the apron. |
| 11. To illuminate | - To light up, to make visible in the dark. |
| 12. Wand | - Illuminated sticks used by the marshaller. |
| 13. To cut engines | - To stop engines. |

CONVERSATION

- Can I control the brightness or intensity of the lights in this room? Who controls the intensity of the lights at an aerodrome? Is the intensity of the lights high in thick weather? Could high intensity lights momentarily blind the pilot? When lights momentarily blind the pilot, what does he ask the controller to do?
- A beacon is a signal light. Is an aerodrome beacon also a light? Are there many beacons at an aerodrome? Are aerodrome beacons lit at night? During the day? Are aerodrome beacons lit every day? Is an aerodrome beacon of high intensity or low intensity? Can you see the aerodrome beacon from the road? Who sees the aerodrome beacon? Is it necessary for the intensity of the light of the aerodrome beacon to be high? Is the pilot far from or close to the aerodrome when he first sees the aerodrome beacon?
- Does a wheel rotate? Does the steering wheel in a car rotate? What wheel rotates? Does a pilot distinguish an aerodrome beacon because it rotates? Do runway lights rotate? Taxiway lights? What light rotates?
- Are signals flashes of lights? Is a continuous light a flash? What do you do to have flashes of light?
- Does an identification beacon also flash a coloured light? Can the flashes of an identification beacon be seen by the pilot? By the radar controller? Who sees the flashes of the identification beacon? By night? By day? Why is it necessary for the pilot to see the identification beacon?
- Morse code is a signal. Can we signal by morse code with sounds? By light? By telegraph? Who uses morse code with light signals? What lamp does the controller use when he signals by morse code? What beacon identifies the aerodrome with morse code?

7. Are the runway edges on both sides of the runway? Are there two runway edges on each runway? Three? How many runway edges are there? Where are the runway edges? Why do runway edges have lights? Markings? Are the runway edge markings long or short?
8. When a pilot approaches an aerodrome does he align his plane with the runway? With the perimeter? With the boundary? With what does he align his plane? When does a pilot align his plane with the runway?
9. VASIS is a system which projects a beam of light. Is VASIS a glide path formed by lights? Markings? Electronics? What sort of glide path is VASIS? Is VASIS used as a landing aid? To take off? When is VASIS used? How many different colours of lights does VASIS have?
10. Is a marshaller the person who directs aircraft on the apron? Does the marshaller signal the pilot? Does the marshaller signal him to stop? To come forward? Turn left? Turn right? Where is the marshaller when he signals the pilot?
11. If I want to illuminate this room, I put on the lights. Do people illuminate buildings? Trees? Do they use one light or many to illuminate a park? An aerodrome? At what time do they illuminate this city? This aerodrome?
12. At night wands are used by the marshaller. How many wands does the marshaller have? Does the marshaller wave the wands? Does he point the wands to the left? Right? Up? Down? Are these wands illuminated? By night? By day? Does a marshaller always use wands? When does the marshaller use wands? Why does the marshaller use wands?
13. Does the marshaller signal the pilot to cut his engines? When the pilot cuts the engines, do the engines still function? Do the engines continue to make a noise when they are cut? What happens when the pilot cuts the engines? Does the marshaller cut the engine? Who cuts the engines? Where is the marshaller when he asks the pilot to cut the engines?



WRITTEN EXERCISE

Instructions to students: Choose the appropriate meaning or meanings, if more than one meaning applies. Mark the square accordingly.

A vector is

- | | | |
|-----|------------------------------------|--------------------------|
| a.) | a navaid | ☐ |
| b.) | a landing aid. | ☐ |
| c.) | a beacon. | ☐ |
| d.) | a heading given by the controller. | ☐ |

A controller can

- | | | |
|-----|---|--------------------------|
| a.) | vary the intensity of the aerodrome beacon. | ☐ |
| b.) | vary the intensity of the runway lights. | ☐ |
| c.) | signal in morse code. | ☐ |
| d.) | turn on the lights of the runway in use. | ☐ |

Aerodrome markings

- | | | |
|-----|----------------------------------|--------------------------|
| a.) | are on the runway threshold. | ☐ |
| b.) | are on the taxiways. | ☐ |
| c.) | can be seen at a great distance. | ☐ |
| d.) | can be seen in poor visibility. | ☐ |

Approach lights

- a.) permit a pilot to align his aircraft ☐
- b.) with the runway. ☐
- c.) are seen below the glide path. ☐
- d.) can be seen at a great distance. ☐
- e.) are seen just before touchdown. ☐

A beacon is

- a.) a row of lights. ☐
- b.) a light of high intensity. ☐
- c.) like a search light. ☐
- d.) the light from a flash light. ☐

A beam of light

- a.) can illuminate all angles at the same time ☐
- b.) is confined to one area at a time. ☐
- c.) is a morse code signal. ☐
- d.) if directed at you, can be momentarily blinding. ☐

A red light indicates

- a.) hazards. ☐
- b.) danger areas. ☐
- c.) obstacles. ☐
- d.) to the pilot he is below the approach slope. ☐

VASIS is

- a.) a landing aid. ☐
- b.) distinguishable by its green light. ☐
- c.) an electronic navaid. ☐
- d.) comprised of red and white lights. ☐

LESSON 13

A SHORT HISTORY OF RADAR

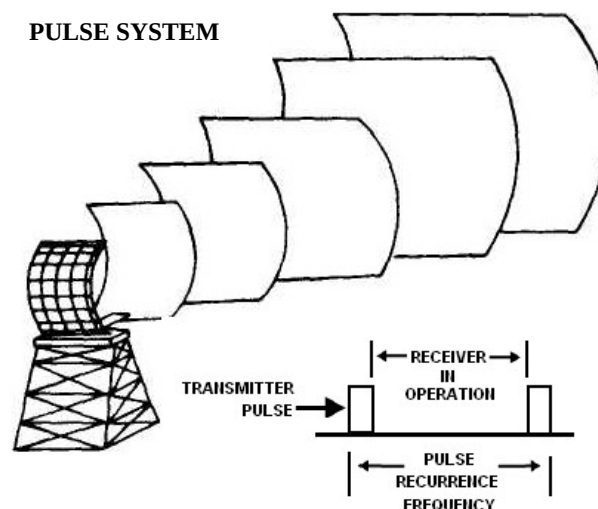
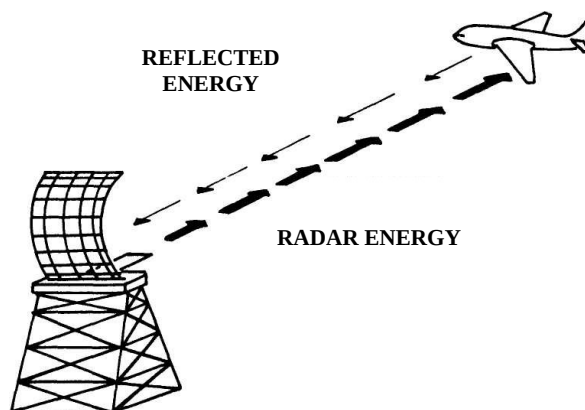
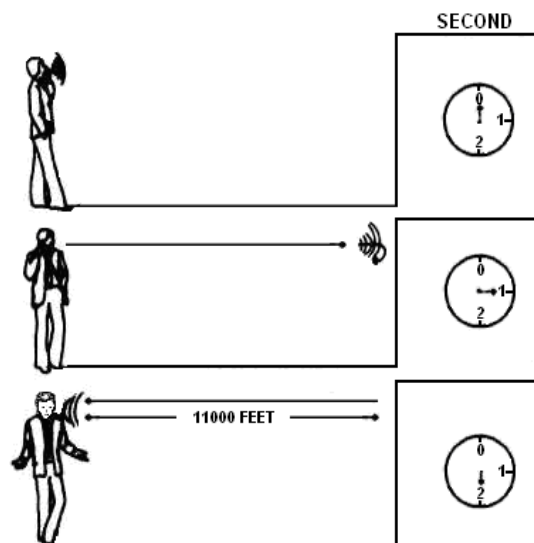
The principles of radar are not new; in fact, some early experiments were made back in the 1880s. In 1904 a German engineer had invented, as he explained, a radio-echo (1) "collision prevention device". By 1922 the famed electronics engineer, Marconi, devised a principle for sending radio signals between ships which would be reflected (2) back to a receiver on the sending ship and thereby immediately reveal the presence and bearing (3) of the other ship in fog or thick weather. This is the basic principle used in all radar.

Further improvements on this concept were developed, including the introduction of the "pulse" (4) principles on which modern radar is based. During the period 1935 - 1940, successful pulse radar systems were produced independently and, nearly at the same time, in the United States, England, France and Germany.

The application of radar in the air traffic control system consists of two basic designs. The initial type of radar, called primary radar, (5) began to be used in most parts of the world in the early 1950s. Another form of radar, secondary surveillance

(SSR) (6) is used for advanced air traffic control. When the word "radar" is used alone, it usually includes both primary and secondary radar.

In primary radar a beam of individual pulses of energy is transmitted from the ground equipment at the rate of approximately 1,200 per second, while the transmitting antenna rotates at a speed of 3 to 6 rpm (7) for long-range systems, and as fast as 60 rpm for airport coverage. These pulses hit the aircraft from 16 to 34 times each scan, (8) depending upon the rotation rate of the antenna and the width of the beam. An aircraft in the path of this radar beam will reflect back some of the pulses which are picked up by a receiving element on the ground antenna. The strength of the reflected energy depends on the aircraft's size and attitude, in addition to the power of the transmitter. This reflected energy produces a bright "echo" or "target" (9) on a cathode ray tube. (10) The bearing of the target from the radar site is known from the location of the echo on the CRT. The distance of the target from the radar site is determined in the time it takes the radar pulse to travel from the radar site to the aircraft and back (about 1 nautical mile in 12.34 millionths of a second or micro seconds).



The most common type of primary radar is the terminal area surveillance radar (TAR) **(11)** which was designed as a medium range radar - about 50 miles – for the control of traffic in the vicinity of an airport. While the progress of moving targets is constantly tracked, **(12)** display of fixed echoes reflecting from mountains or other obstructions may or may not be displayed. The normal rotation of a TAR is approximately 13 rpm, so that the traffic situation is updated every few seconds.

Another type of primary radar used in the control of air traffic is the long-range radar known as the en-route surveillance radar (RSR). **(13)** This system has a range up to about 200 miles and will detect aircraft up to an altitude of about 40,000 feet. It is used in area control centres for the control of en-route traffic. The RSR normally is provided with features similar to the TAR. Because of its slower rotation – 3 or 6 rpm and other factors, its accuracy and resolution are not as high as the TAR.

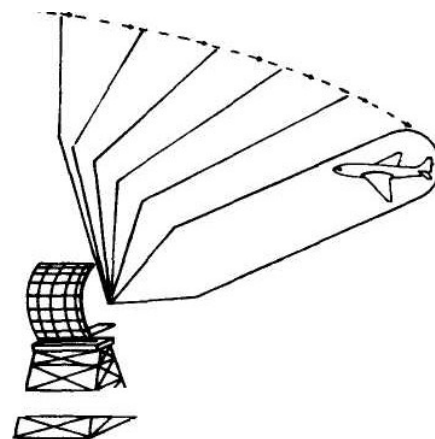
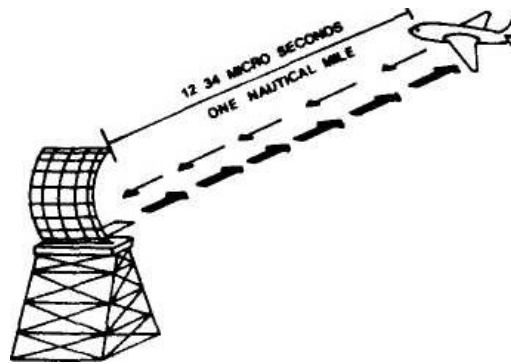
These are the more common types of primary radar used today; other primary radar are, PAR **(14)** and GCA **(15)** which are not so common.

VOCABULARY

- | | |
|--|--|
| 1. Echo | - The aircraft as seen on the radar scope. Also called blip or target (see below). |
| 2. To reflect | - To return heat, light or sound. In this lesson it is the radio signal that is returned or reflected back to a receiver. |
| 3. Bearing | - The angle between a direction (object, radio emission, etc.) and a reference direction as determined at the place of the observer. |
| 4. Pulse | - A single beat or vibration of sound, light, radio or radar. |
| 5. Primary radar | - A radio detection device which, by use of reflected radio signals, provides information on range and azimuth of objects. |
| 6. Secondary surveillance radar | - A radar system wherein a radio signal transmitted by a ground interrogator initiates the transmission of a radio signal from an airborne transponder. With secondary surveillance radar additional information may be fed into the radar scope such as the identification of the aircraft, its altitude and ground speed in knots. |
| 7. rpm | - Revolutions per minute. |
| 8. To scan; the scanner | - To sweep or to search by means of radar. The scanner is the aerial assembly which, by rotating, scans the area to be searched. The scan is the movement of the scanner. |
| 9. Target | - Meaning the aircraft as seen in the radar scope. It is also called echo or blip (see above). |
| 10. Cathode Ray Tube (CRT) | - Another word for radar scope or radar display or PPI. |
| 11. Terminal Area Surveillance Radar (TAR) | - Aerodrome radar. Also called airport surveillance radar. |
| 12. To track | - To follow the movement of an aircraft. |
| 13. En-Route Surveillance Radar (RSR) | - Also called area / airways radar. |
| 14. PAR | - Precision Approach Radar. |
| 15. GCA | - Ground Controlled Approach. |

CONVERSATION

1. Is the echo an aircraft as seen on the scope? What does an echo represent on the radar scope? What is another word for echo? And another?
2. Is your face reflected in a mirror? Can light be reflected? Can a ball be reflected? What can be reflected?
3. Can a compass show you your bearing? Can an altimeter show your bearing? What shows you your bearing? Who is interested in knowing the bearing of an aircraft at an aerodrome?
4. If I touch my wrist, can I feel my pulse? When I run, my pulse is fast? Does modern radar have a pulse? Is the pulse principle important in modern radar? Is the pulse principle a new concept in radar? At what period in history was the pulse principle introduced into radar?
5. Was primary radar the first type of radar? Is primary radar still used in civil aviation? When was primary radar first used in aviation? Do you only see the aircraft with primary radar? Do you only see a mountain with primary radar? Is primary radar the most advanced type of radar? What type of radar is primary radar?
6. When we talk about radar do we mean primary and secondary? Is secondary surveillance radar a more advanced type of radar than primary? Does secondary surveillance radar only show the aircraft on the scope? What does secondary radar show on the scope? Tell me the difference between primary and secondary radar? What does SSR stand for?
7. Rpm is an abbreviation for revolutions per minute? Are 6 rpm's faster than 3 rpm's? Does a pilot watch the rpm's of the engine? If the rpm's of his engine slow up, what is happening to his engine? Does an antenna rotate at a speed of 3 rpm's?
8. When I scan a page in this book I look quickly at the page in order to find something. Did I scan the page? Did I read every word? What did I do? Will you read every word in this sentence? Why didn't you scan the sentence? Because you read every word. When would you scan a page? Do you look quickly at the page when you scan it? Does a radar antenna scan an aircraft slowly or quickly? Is a scanner an antenna? Does the scanner rotate? Does a scanner rotate in the direction of the aircraft? Who uses the scanner? Why does a controller use the scanner? Is a scan a movement of the scanner? Are there pulses in each scan? Do pulses hit an aircraft in each scan? Can pulses hit an aircraft 16 times in each scan? 34 times? How many pulses are there in each scan?
9. Can a target be seen on a radar scope? Is a target an aircraft? Does the scanner pick up a target? Is a mountain a target? What is a target? What is another word for target?
10. Is a Cathode Ray Tube another word for radar scope? Is the Cathode Ray Tube in the control tower? Are there many Cathode Ray Tubes in the control tower? Does a controller see the targets on the Cathode Ray Tube? What are other words for Cathode Ray Tube? What is the abbreviation for Cathode Ray Tube?
11. Is terminal area surveillance radar a primary type radar? Is terminal area surveillance radar at most airports? Does terminal area surveillance radar control the traffic at the airport? Does terminal area surveillance radar control the traffic en-route? In the airways? What traffic does terminal area surveillance radar control? What is the range of TAR?
12. A controller who follows an aircraft on his radar scope tracks an aircraft. Does he track an aircraft on his radar scope? Does he track mountainous terrain? Birds? Clouds? What does he track on his radar scope? On what does a controller track an aircraft?
13. Is en-route surveillance radar known as RSR? Is en-route surveillance radar known as long range radar? Airways? Area? Aerodrome? What is en-route surveillance radar known as? What controller tracks an aircraft with RSR? Is RSR at all reporting points? What can a controller track with RSR?
14. Is PAR an abbreviation for Precision Approach Radar? What is PAR short for?
15. Is GCA short for Ground Controlled Approach? Is GCA similar to PAR? What is GCA short for?



WRITTEN EXERCISE

Instructions to students: Answer the questions.

I) What is the difference between the position and bearing of an aircraft?

II) What light/s or lamp/s radiate a beam?

- | | | |
|-------|----------------------|--------------------------|
| (i) | an Aldis lamp | ☐ |
| (ii) | Crossbar lights | ☐ |
| (iii) | Portable signal lamp | ☐ |
| (iv) | Aerodrome beacon | ☐ |

(a) Reflection comes from different sources; name three, e.g. a mirror is one.

(b) Name three types of radar.

(c) Why does an airliner always fly IFR?

(d) An aircraft seen on a Cathode Ray Tube is called what?

(e) A Cathode Ray Tube is also known as what? •

(f) Does a scan hit an aircraft once or many times?

(g) Where is the radar antenna at this aerodrome?

LESSON 14

SOME PROBLEMS ASSOCIATED WITH RADAR

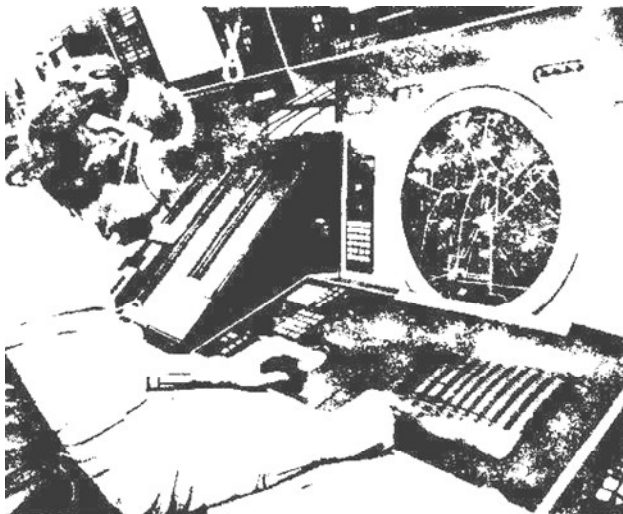
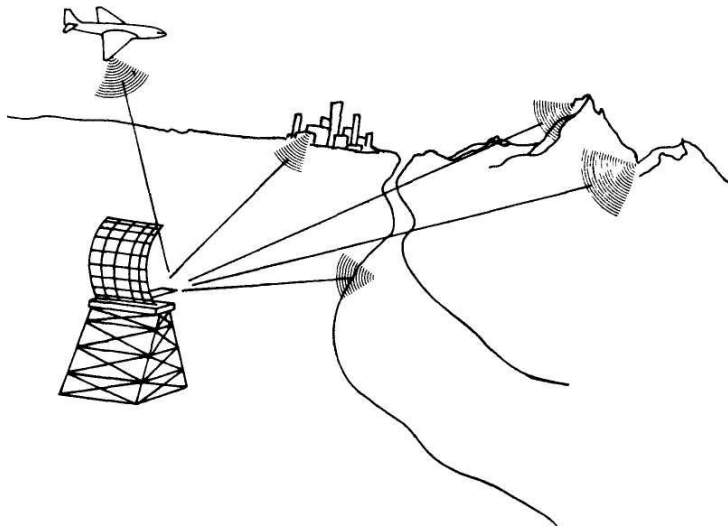
One of the problems associated with radar is the amount of unwanted clutter (1) on the radar scope which the scanner picks up. This unwanted clutter is mainly echoes of stationary objects, such as mountains and other elevated terrain, as well as other objects situated close to the transmitting/receiving antennae.

With too much clutter, a controller cannot always see the aircraft. MTI (2) allows him to reduce such clutter. The MTI does this by not allowing echoes received from stationary objects to be displayed on the radar scope. A controller has an MTI switch at his console and with this he can control, to a degree, the amount of clutter caused by stationary objects.

However, clutter is not caused by stationary objects only, echoes received from precipitation and/or clouds also cause clutter. MTI is effective in removing clutter caused by precipitation and clouds to a limited extent since MTI permits any echo which is moving to be displayed. So an additional technique known as Circular Polarization (3) has been developed which prevents the display of most of the clutter caused by precipitation and clouds.

While using Circular Polarization, however, controllers would not be aware of the existence of any weather lying ahead of the aircraft, since all echoes from precipitation and clouds would be removed from his scope. Because of this, he could conceivably vector a pilot into turbulent or hazardous (4) meteorological conditions.

A solution to this problem has been to show on the scope the actual weather in the area. This clutter can be switched on and off. Another solution is to feed the weather information into the automatic data processing (5) equipment for incorporation into the controller's computer processed radar scope. This information appears in digital form on the radar scope.



In most radar equipment, the controller is provided with the capability of using MTI alone, MTI plus Circular Polarization, or Circular Polarization alone, thus allowing him a considerable amount of flexibility. (6)

Besides the problems caused by clutter, the slashes (7) on the SSR scope, if too large, could overlap (8) with other slashes making it difficult to distinguish individual aircraft. To keep large slashes apart would require the controller to provide extensive separation between aircraft with resulting delays. This too is improved somewhat with the help of radar automation.

Although most of the deficiencies (9) in radar are characteristic of radar technology, improvements are constantly being made which will assist the air traffic controllers in the future.

VOCABULARY

- | | |
|------------------------------|--|
| 1. Clutter | - The unwanted "snow" or interference on a radar scope caused by fixed or moving objects. |
| 2. Moving Target | - With MTI a controller can switch off clutter caused by fixed objects such as mountains, buildings, etc. |
| 3. Circular | - With Circular Polarization a controller can switch off clutter caused by very slow moving objects, such as clouds and precipitation. |
| 4. Hazardous | - Dangerous. |
| 5. Automatic data processing | - Another word for computer. |
| 6. Flexibility | - The ability to change easily. |
| 7. Slashes | - A return or echo from secondary surveillance radar (not primary radar which is a blip). |
| 8. Overlap | - One item placed over another item, either partly or completely. |
| 9. Deficiencies | - Incomplete, lacking perfection. |

CONVERSATION

1. Does a controller see clutter on his radar scope? Do mountains cause clutter on a radar scope? Buildings? Clouds? An aircraft? What is clutter? Can clutter move? Is it easy to see the target with clutter? What type of radar has clutter?
2. Can a controller switch off clutter? With MTI can he switch off stationary clutter? Fixed clutter? With what does he switch off stationary clutter?
3. Can a controller switch off clutter caused by moving objects with Circular Polarization? Stationary objects? What can he switch off with Circular Polarization? Can he switch off the moving target with Circular Polarization? What does he switch off with Circular Polarization?
4. Are hazardous conditions dangerous? Are low clouds around an airport hazardous? Obstructions on the runway? Passengers smoking at take-off? A meal from the galley? What do you consider hazardous in aviation?
5. Is automatic data processing a modern invention? Do we use automatic data processing in aviation? In other fields? Did Marconi (the inventor) have automatic data processing? Who uses automatic data processing?
6. Does a controller have flexibility with Circular Polarization and MTI? Does this flexibility permit him to use Circular Polarization and MTI? Circular Polarization or MTI? Does this flexibility permit him to use Circular Polarization alone? Does this flexibility permit him to show only clouds? Only precipitation? Or both?
7. Are slashes another term for target? Are slashes another term for blip? Echo? Are slashes a description of a dot or a line? Are slashes seen on SSR? Primary radar? On what type of radar are slashes seen?
8. Do two aircraft on the scope appear as one slash when they overlap? Do slashes always overlap on the scope? Sometimes? Partly? Completely? Is it good that slashes overlap on the scope? Why is it necessary to separate them when they overlap?
9. Does everybody have some deficiencies in their character? Is good humour a deficiency? Good manners? What are one's deficiencies? Are there deficiencies in aviation? Can automation help to eliminate some of the deficiencies in aviation? Data processing? Computers? New inventions? Tell me some of the deficiencies inherent in radar.

WRITTEN EXERCISE

Instructions to students: Describe in your own words, "The causes and effect of clutter and how a controller handles it".

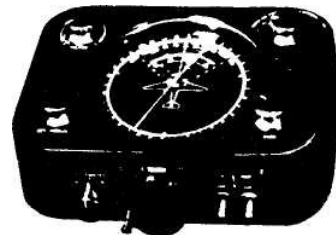
LESSON 15

OTHER NAVIGATIONAL AIDS

We have seen, in previous lessons, how VOR/DME/ILS work. However, we have not yet discussed Direction Finding (1) which is one of the oldest methods of navigation. Direction finding originally was developed in the 1920s to provide a "fix" (2) on an aircraft's position. Generally speaking, this is an emergency type of navigational device (3) which enables a controller to help a "lost" pilot find an airport by giving him headings to follow. The technique is useful on occasions in search and rescue operations to plot the location of an aircraft at an unknown position.

Automatic direction finding (ADF) is an airborne counterpart (4, 5, 6) of the ground Direction Finding equipment. In an aircraft, the ADF permits a pilot to take continuous bearings automatically on any ground station operating on a frequency usable by the airborne equipment. The ground stations which are available to the pilot for this purpose include "compass locators", Non-directional Radio Beacon (7, 8) (NDB), and commercial radio-broadcasting stations. All of these may be used by the ADF-equipped aircraft for en-route navigation and for approaches to certain airports in remote areas. ADF is used mainly as a "backup" (9) navigational device, or in areas of the world where traffic is light and ground navigation aids are limited. It does not provide the position-fixing accuracy or display needed for modern air navigation in high-density (10) traffic patterns.

There are other navigational aids: Decca and LORAN (11, 12) are two of the most commonly known in aviation.



Direction Finder

VOCABULARY

- | | | | |
|-----|-------------------|---|---|
| 1. | Direction Finding | - | A navigational aid. |
| 2. | Fix | - | Position of an aircraft as determined by various means of navigation. |
| 3. | Device | - | An instrument or equipment designed for a particular purpose. |
| 4. | ADF | - | Automatic Direction Finding. A navigational aid in the aircraft. |
| 5. | Airborne | - | An aircraft in flight is airborne. In this particular case, however, it means equipment installed in the aircraft, known as airborne equipment. |
| 6. | Counterpart | - | Equivalent. |
| 7. | Compass locators | - | Low powered radio beacons frequently installed at the same locations as outer and middle markers. |
| 8. | NDB | - | Non-directional radio beacon. |
| 9. | Backup | - | Support or help. |
| 10. | High density area | - | An area with a lot of traffic. |
| 11. | Decca | - | A navigational aid. |
| 12. | LORAN | - | A navigational aid, known as Long Range Navigation. |

CONVERSATION

1. Is Direction Finding an old method of navigation? Was Direction Finding developed in the 1920s? The 1930 s? When was Direction Finding developed? Is Direction Finding still a navigational aid today? Is the technique of Direction Finding still used today? Is Direction Finding used for search and rescue operations? Is Direction Finding used to plot the location of an aircraft? Why is Direction Finding a useful method of navigation?
2. Does a navigator want a fix on his position? A pilot? A radio technician? Who wants a fix on his position? Does Direction Finding enable a pilot to get a fix on his position? VOR? DME? MTI? What enables a pilot to get a fix on his position? Is a pilot lost when he has a fix on his position? Does he call MAYDAY when he has a fix on his position? Does he need search and rescue operations when he has a fix on his position? When does he want a fix on his position? What enables a pilot to get a fix on his position?
3. Is a compass a useful device? A tachometer? A spanner? Was the wheel the first useful device ever invented? What do you consider was the first useful device ever invented? Name a useful navigational device.
4. Is ADF also a navigational aid? Is ADF in the aircraft? Can the pilot get his bearings with ADF? Can a pilot get his bearings automatically with ADF? Can a pilot get a fix on his position with ADF? Is ADF also used by controllers? Who uses ADF? Why does a pilot use ADF?
5. If ADF is in the aircraft, is it airborne? Is all equipment on the flight deck airborne equipment? Is a pilot always airborne? A stewardess? When is a pilot airborne? Why is equipment called airborne equipment? Why is ground equipment not called airborne equipment?
6. Is ADF similar equipment to Direction Finding equipment? Is ADF a useful counterpart to Direction Finding equipment? Is a windsock a useful counterpart to a wind direction indicator? Are some grammar books useful counterparts to other grammar books? Is this book a counterpart to other aviation language books? To a grammar book? What is a useful counterpart to this book? To ADF? To a windsock?
7. Is the compass locator situated at the aerodrome? Does the compass locator help the pilot get a fix on his position? Is a compass locator one of many devices enabling a pilot to get a fix on his position? Where is a compass locator situated? Is the compass locator a radio beacon? Is the compass locator a high frequency radio beacon? What sort of radio beacon is a compass locator?
8. Is NDB a non-directional radio beacon? Is NDB a useful counterpart to other navigational aids? Is NDB situated at the aerodrome? Can a pilot take his bearings from NDB? Is NDB airborne equipment? Where is NDB? What does NDB stand for?
9. Do you help friends when they are in trouble? Do you backup friends when they are in trouble? When they need you? Is ADF used as backup equipment to other navigational equipment? Why is it necessary to have backup equipment in navigation?
10. Does Chicago International Aerodrome have the highest density traffic in the world? Does Charles de Gaulle International have high density traffic? Heathrow International? What aerodrome has the highest density traffic?
11. Is Decca a navigational aid? Is Decca one of many navigational aids? Is Decca used on the North Atlantic? Pacific? Everywhere in the world? Where is Decca used?
12. Is LORAN also a navigational aid? Is LORAN a navigational aid approved by ICAO? Do pilots also approve of LORAN as a navigational aid? What does LORAN stand for?

WRITTEN EXERCISE

Instructions to students: Choose the appropriate meaning, or meanings, if more than one meaning applies. Mark the square accordingly.

a) Every aircraft has, on its flight deck

- i. airborne equipment. ☐
- ii. a compass. ☐
- iii. a tachometer. ☐
- iv. ADF. ☐

b) A pilot cannot navigate without

- i. ICAO. ☐
- ii. finding his bearings. ☐
- iii. having a fix on his position ☐
- iv. Decca. ☐

c) Search and rescue operate

- i) when they receive a MAYDAY signal. ☐
- ii) when they receive a distress signal. ☐
- iii) in an emergency. ☐
- iv) with Direction Finding equipment. ☐

d) With Direction Finding equipment a controller can

- i) vector a pilot into position. ☐
- ii) help a pilot who is lost. ☐
- iii) give a pilot headings to follow. ☐
- iv) direct an aircraft through high density traffic. ☐

a) Navaid equipment can consist of ☐

- i) LORAN. ☐
- ii) Decca. ☐
- iii) commercial radio broadcasting stations ☐

b) When approaching an aerodrome a pilot sees

- i) an NDB beacon. ☐
- ii) an aerodrome beacon. ☐
- iii) a morse code signal. ☐
- iv) a compass locator. ☐

c) In an emergency it is useful for a pilot to have

- i) backup equipment. ☐
- ii) Direction Finding equipment. ☐
- iii) high density fog ☐
- iv) a fix on his position. ☐

d) Some airborne equipment is

- i) ADF. ☐
- ii) ILS. ☐
- iii) VASIS. ☐
- iv) a transponder. ☐

Instructions to students: Describe in your own words the functions of the following, distinguishing between a navaid and a landing aid:

VASIS

ADF

ILS

NDB

DME

VOR

VOCABULARY INDEX

A

aboard (1)
advice (3)
advise, to (3)
aerodrome beacon (12)
Aerodrome Surveillance Radar (ASR) (13)
Aeronautical Information Publication (AIP) (2)
affect, to (3)
airborne (15)
aircraft identification (7)
airway (2)
aldis lamp (4)
align, to (12)
alternate aerodrome (2)
altimeter setting indicator(4)
area of high pressure (3)
area of low pressure (3)
automatic data processing (14)
Automatic Direction Finding(ADF) (15)
Automatic Terminal Information Service (ATIS) (4)

B

back-up (15)
barometer (10)
base leg (4)
beam (9)
bearing (8)
briefing officer (2)

C

Cathode Ray Tube (CRT) (13)
ceiling (3)
centre line (2)
Circular Polarization (14)
clearance (1)
clear of runway (1)
clutter (14)
cold front (3)
compass locators (15)
computer (7)
console (4)
contact, to (1)
co-ordinate, to (5)
counterpart (15)
current flight data (7)
current weather conditions (3)
cut engines, to (12)
cruising speed (2)

D

Decca (15)
deficiencies (14)
device (15)
dew point (10)
direction finding (15)

Note: () Indicates Lesson Number

Distance Measuring Equipment (DME) (8)
domestic flights (6)
downwind leg (4)

E

echo (13)
effect (3)
elevation of aerodrome (2)
emergency procedures (1)
Enroute Surveillance Radar (RSR) (13)
Expected Approach Time (EAT) (5)
Expected Time of Arrival (ETA) (10)

F

facilities (2)
fan marker (9)
fasten, to (1)
filed true airspeed (7)
final approach (4)
fix (10) (15)
flashes (12)
flexibility (14)
flight attendant (1)
flight level (1)
flight plan (2)
flight progress board (7)
flight progress strips (7)
front (3)
full-load (10)
functions (7)

G

glide path equipment (9)
go ahead (1)
Ground Control Approach (GCA) (13)
ground speed (3)

H

hazardous (14)
heading (1)
headset (4)
headwind (3)
hectic (10)
high density area (15)
high pressure area (3)
holding pattern (10)
holding point (6)
holding position (I)
home-on, to (10)

I

inbound (10)
identification beacon (12)
illuminate, to (12)
inner marker (9)
Instrument Flight Rules (IFR) (2)
Instrument Landing System (ILS) (9)
Instrument Landing System indicator (9)
Instrument Meteorological Conditions

(IMC) (3)
intercept, to (10)
interference (8)
international airport (6)
International Civil Aviation
Organization (ICAO) (9)
intensity (12)
interrogator (8)

K

knots (10)

L

landing sequence (4)
lateral separation (7)
local flights (10)
localizer (9)
longitudinal separation (7)
long range radar (13)
Loran (15)
loudspeaker (4)
low pressure area (3)

M

maintain, to (1)
manoeuvre, to (4)
manually (7)
marshaller (12)
meteorologist (3)
microphone (4)
morse code (12)
Moving Target Indicator (MTI) (14)

N

nautical miles (8)
navaids (8)
Non-Directional Radio Beacon (NDB) (15)
Notices to Airmen (NOTAM) (2)

O

observe, to (1)
outbound (10)
outer marker (9)
over (1)
overcast (3)
overlap (14)
overshoot (9)

P

precipitation (3)
precise (9)
Precision Approach Radar (PAR) (13)
primary radar (13)
procedure turn (10)
pulse (13)

R

radials (8)
radio navigation aids (navaids) (8)
ramp (1)
reading (8)
reflect, to (13)
relay, to (1)
reporting points (2)

restricted area (4)
revert back, to (6)
revolutions per minute (rpm) (13)
roger (1)
rotate, to (12)
routine (7)
runway edge (12)
runway in use (4) light gun (4)

S

scanner (13)
Secondary Surveillance Radar (SSR) (13)
sector (7)
signalling lamp (4)
slashes (14)
specify, to (7)
spread, to (10)
stack, to (5)
static (8)
stray, to (2)
subsonic aircraft (6)
supersonic aircraft (6)
synoptic chart (3) middle marker (9)

T

tailwind (3)
target (13)
Terminal Area Surveillance Radar (TAR) (13)
threshold (9)
tower cab (4)
track, to (13)
traffic circuit (4)
transponder (8)
true airspeed (7)
tune in to, to (1)
type of aircraft (6)

U

Ultra High Frequency (UHF) (5)
unaware of (1)

V

vector, to (10)
vertical separation (7)
Very High Frequency (VHF) (5)
Very High Frequency Omni directional Range
(VOR) (8)
Visual Approach Slope Indicator System
(VASIS) (12)
Visual Flight Rules (VFR) (2)
Visual Meteorological Conditions (VMC) (5)

W

wake turbulence (10)
wand (12)
warm front (3)
weather forecast (3)
wind direction indicator (4)
wind speed indicator (4)
workload (7)



Course 291

AVIATION ENGLISH FOR AIR TRAFFIC CONTROLLERS

*For application by Instructors at ICAO-assisted Training Centres and for
the unrestricted use of other interested Organizations*